6. Sunny and Paula together can complete a work in 12 hours while Paula can complete the same work in 18 hours alone. Another person Jane takes 45 hours to complete the same work alone. How many hours will sunny and Jane take to complete the same work?

(a) 15 (b) 16
(c) 18 (d) 20

RRB RPF Constable – 25/01/2019 (Shift-III)

Ans. (d) : Suppose sunny will take time of x hours for completing the work

According to the question,

$$\frac{1}{x}+\frac{1}{18}=\frac{1}{12}$$

$$\frac{1}{x}=\frac{1}{12}-\frac{1}{18}$$

$$\frac{1}{x}=\frac{3-2}{36}=\frac{1}{36}$$

x = 36 hour

work done by sunny and Jane together-

$$\frac{1}{36}+\frac{1}{45}=\frac{5+4}{180}=\frac{9}{180}$$

Taken time in doing work by sunny and Jane $=\frac{180}{9}=20$ day

7. A works four times faster than B. so he can complete a work 60 days earlier than B. In how many days can they complete the work if they work together?

(a) 21 days (b) 22.5 days
(c) 16 days (d) 25 days

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (c) : Let B does a work in x day and A does that work in (x – 60) days. Then on the basis of work efficiency.

x = (x – 60)4
x = 4x – 240
3x = 240
x = 80 days

or If B does any work in 80 days then one the basis of work efficiency A will do that work in 20 days.

one day work of (A + B) $=\frac{1}{80}+\frac{1}{20}$

$$=\frac{1+4}{80}=\frac{5}{80}=\frac{1}{16}\text{ part}$$

So A and B do work together then they can complete work in the 16 days

8. Brij alone can paint a wall in 7.2 days while Madhu takes 10.8 days to do the same work. How many days will they take to paint 5/6 part of the wall when working together?

(a) 4.2 (b) 3.6
(c) 3.9 (d) 4.8

RRB Group-D – 26/09/2018 (Shift-I)

Ans : (b) Let they take t days to paint the 5/6 part of the wall.

According to the question,

$$\Rightarrow \frac{t}{7.2}+\frac{t}{10.8}=\frac{5}{6}$$

$$\Rightarrow \frac{10t}{72}+\frac{10t}{108}=\frac{5}{6}$$

$$\Rightarrow \frac{30t+20t}{216}=\frac{5}{6}$$

$$\Rightarrow \frac{50t}{216}=\frac{5}{6}$$

$$\Rightarrow t=\frac{5\times 216}{6\times 50}$$

$$\Rightarrow t=\frac{36}{10}$$

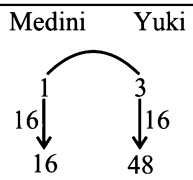
$$\boxed{t=3.6\text{ day}}$$

9. Medini can paint the entire wall alone in 16 days while Yuki takes three times longer than this. In how many days will they paint half of the wall if they work together?

(a) 12 (b) 3
(c) 6 (d) 9

RRB Group-D – 11/10/2018 (Shift-II)

Ans : (c)



One day work of both $=\frac{1}{16}+\frac{1}{48}$

$$=\frac{3+1}{48}=\frac{4}{48}=\frac{1}{12}\text{ part}$$

Time taken in completing the work = 12 days

So time taken for completing half work $=\frac{12}{2}=6$ days

10. A can finish a work in 5 days while B takes 10 days to do the same work. They start working together but A has to leave the work 4 days before the work is over. How many days did A work?

(a) 1 (b) 1.5
(c) 2 (d) 2.5

RRB Group-D – 10/10/2018 (Shift-III)

Ans : (c)

One day work of A $=\frac{1}{5}$ part

One day work of B $=\frac{1}{10}$ part

Let B work for x days

∴ Work done by A = (x – 4) day

According to the question

$$\frac{(x-4)}{5}+\frac{x}{10}=1$$

$$\frac{2(x-4)+x}{10}=1$$
$2x-8+x=10$
$3x-8=10$
$3x=18$
$\boxed{x=6}$
Time taken by A to complete the work = (x – 4)
= 2 day

11. A can finish a work in 5 days and B can finish the same work in 8 days. If both work together and earn Rs. 6760. Find the share of A.
(a) Rs. 4,160 (b) Rs. 3,600
(c) Rs. 5,070 (d) Rs. 4,056
RRB Paramedical Exam – 20/07/2018 (Shift-II)

Ans : (a) One day work of A $=\frac{1}{5}$ part
one day work of B $=\frac{1}{8}$ part
work ability ratio of A and B $=\frac{1}{5}:\frac{1}{8}$
$\boxed{A:B=8:5}$
So share of A $=\frac{6,760}{13}\times 8$ = 520 × 8 = Rs. 4,160

12. A and B take 25 and 45 days respectively to complete a task. A started working alone and after a few days B started working together. The work took a total of 20 days to complete the After how many days did B start working?
(a) 10 (b) 11
(c) 12 (d) 9
RRB Group-D – 30/10/2018 (Shift-III)

Ans. (b) : Let B started work after x day.
According to the question,
$$\frac{20}{25}+\frac{20-x}{45}=1$$
$$\frac{4}{5}+\frac{20-x}{45}=1$$
$$\frac{20-x}{45}=1-\frac{4}{5}$$
$$\frac{20-x}{45}=\frac{1}{5}$$
$20-x=9$
x = 20 – 9
x = 11 days

13. A can do a work in 15 days and B can do it in 25 days. A and B started working together but B left the work 7 days before the work completed. How many days did they work together?
(a) 6 (b) 5
(c) 8 (d) 9
RRB Group-D – 07/12/2018 (Shift-III)

Ans : (b) one day work of A $=\frac{1}{15}$ part
one day work of B $=\frac{1}{25}$ part
one day work of (A+B) $=\left(\frac{1}{15}+\frac{1}{25}\right)=\frac{8}{75}$ part
∵ B left work at 7 day before
Then 7 day work of A $=7\times\frac{1}{15}=\frac{7}{15}$, remain work
$=\left(1-\frac{7}{15}\right)=\frac{8}{15}$ part
Time taken by (A + B) to complete $\frac{8}{15}$ part of work
$=\frac{8}{15}\times\frac{75}{8}=5$ day

14. M can do a work in 12 days. N takes half the time of M to finish the same work. If they both work together, how many work will they do in one day?
(a) 1/3 (b) 1/2
(c) 1/4 (d) 1/5
RRB NTPC 05.04.2016 Shift : 2

Ans : (c) Time taken work by N
$=12\times\frac{1}{2}=6$ day
Work done by both in one day $=\frac{1}{12}+\frac{1}{6}$
$=\frac{1+2}{12}=\frac{3}{12}=\frac{1}{4}$ part

15. A alone can do a work in 15 days and B can do the same work in 18 days. If both work together, how much time will they take to complete the same work?
(a) 10/3 days (b) 36/5 days
(c) 5/36 days (d) 90/11 days
RRB NTPC 02.04.2016 Shift : 3

Ans : (d) Time taken by both to complete the work
$=\frac{15\times 18}{15+18}$
$=\frac{15\times 18}{33}=\frac{15\times 6}{11}=\frac{90}{11}$ days

16. L and M can complete a work in 30 and 90 days respectively. If they work together, in how many days will they complete that work?
(a) 60 (b) 45
(c) 70 (d) $22\frac{1}{2}$
RRB NTPC 31.03.2016 Shift : 1

Ans :(d) Time taken to completing the work by L and M together
$=\frac{30\times 90}{30+90}=\frac{30\times 90}{120}=\frac{45}{2}=22\frac{1}{2}$ days

17. P can do a work in 10 days. Q can do the same work in 15 days. If they work together for 5 days, how much of the work will they complete?

(a) 1/2 (b) 2/3
(c) 1/3 (d) 5/6

RRB NTPC 30.03.2016 Shift : 1

Ans : (d) Work done by both P and Q in one day.

$$=\frac{1}{10}+\frac{1}{15}=\frac{3+2}{30}=\frac{5}{30}=\frac{1}{6}\text{ part}$$

So work done by both in 5 days $=\frac{5}{6}$

18. A can do a work in 10 days and B can do the same work in 20 days. If both of them work together for 2 days, then what part of the work will be finished?

(a) 3/10 (b) 3/8
(c) 1/3 (d) 1/4

RRB NTPC 30.04.2016 Shift : 2

Ans : (a) Work done by A in 1 day $A=\frac{1}{10}$ part

Work done by B in 2 days $=\frac{1}{20}$ part

Work done by both in 2 days $=2\left(\frac{1}{10}+\frac{1}{20}\right)$

$$=2\left(\frac{2+1}{20}\right)=2\left(\frac{3}{20}\right)=\frac{3}{10}\text{ part}$$

Type - 8

19. X and Y can finish a work in 6 days. If X alone can finish the same work in 9 days, then how many days will Y alone take to finish the same work?

(a) 15 (b) 12
(c) 18 (d) 21

RRB RPF Constable – 20/01/2019 (Shift-II)

Ans : (c) one day work of X and Y $=\frac{1}{6}$ part

one day work of X $=\frac{1}{9}$ part

$\therefore$ one day work of Y $=\frac{1}{6}-\frac{1}{9}=\frac{3-2}{18}=\frac{1}{18}$ part

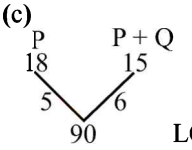
So time taken to complete work by Y = 18 days

20. P alone can complete a task in 18 days. P & Q working together can complete it in 15 days. They started working together. Q left the task after 10 days and P alone had to complete that task. How many days will P take to complete the remaining task.

(a) 5 days (b) 7 days
(c) 6 days (d) 8 days

RRB JE - 22/05/2019 (Shift-III)

Ans : (c)



LCM of 15 and 18 = 90

$\because$ P does 5 unit work in a day and P + Q does 6 unit work in a day

P + Q = 6

5 + Q = 6

Q = 1

So Q will do 1 unit work in a day

10 days work of (P + Q) = 10 × 6 = 60 unit

Remain work = 90 – 60 = 30 unit

P will do the remaining task $=\frac{30}{5}=6$ days

21. P can do any work in 4 hours. Q and R together can complete the same work in 3 hours, while P and R can complete it in 2 hours. If Q alone does this work then how much time will he take to complete this work?

(a) 12 hours (b) 8 hours
(c) 10 hours (d) 15 hours

RRB JE - 23/05/2019 (Shift-II)

Ans : (a) 1 hour's work of P $=\frac{1}{4}$ part .....(i)

1 hour's work of (Q + R) $\frac{1}{3}$ part .....(ii)

1 hour's works of (P + R) $\frac{1}{2}$ part .....(iii)

from the equation (i) and (iii)

$$P+R=\frac{1}{2}\Rightarrow\frac{1}{4}+R=\frac{1}{2}\Rightarrow R=\frac{1}{2}-\frac{1}{4}=\frac{2-1}{4}=\frac{1}{4}$$

$\therefore$ from the equation (ii)

$$Q+R=\frac{1}{3}\Rightarrow Q=\frac{1}{3}-\frac{1}{4}=\frac{1}{12}$$

So taken time to complete the work by Q = 12 hours

22. P is 30% more efficient than Q. If P alone can do a work in 20 days, then Q alone will do this work in how many days?

(a) 18 days (b) 30 days
(c) 26 days (d) 28 days

RRB JE - 23/05/2019 (Shift-III)

Ans : (c) From $E_1D_1 = E_2D_2$ [E = energy, D = Day]

$130 \times 20 = D_2 \times 100$

$D_2 = 26$

So alone Q will do this work in 26 days

23. Srinivas can complete a work in 15 days and Ramesh can do the same work in 9 days, with Ravi's help they completed the work in 3 days. In how many days can Ravi do the same work alone?

(a) $6\frac{3}{7}$ days (b) 10 days

(c) $6\frac{2}{5}$ days (d) $6\frac{1}{5}$ days

RRB JE - 29/05/2019 (Shift-III)

Ans : (a) Let Ravi will complete the work in x day
According to the question,

$$\frac{1}{x}=\frac{1}{3}-\frac{1}{15}-\frac{1}{9}$$

$$\frac{1}{x}=\frac{15-3-5}{45}=\frac{15-8}{45}$$

$$\frac{1}{x}=\frac{7}{45}$$

$$x=\frac{45}{7}=6\frac{3}{7}\text{ days}$$

So Ravi will complete the work in $6\frac{3}{7}$ days

24. After completing a task together in 2 days, P and Q got Rs. 1200 and Rs. 400 respectively. In how many days alone Q will complete the work?

(a) 6 days (b) 3 days
(c) 1/3 days (d) 8 days

RRB RPF SI – 11/01/2019 (Shift-I)

Ans : (d) P : Q
1200 : 400
3 : 1
Q works = (3 + 1) × 2
work of Q = 4 × 2 = 8 days

25. P and Q together can do a work in 7 days. P is twice as efficient as Q. In how many days can P alone do that work?

(a) 21 days (b) 15 days
(c) 10.5 days (d) 12 days

RRB JE - 31/05/2019 (Shift-I)

Ans : (c) Let P does work in x day and Q does in 2x day
According to the question,

$$\frac{1}{x}+\frac{1}{2x}=\frac{1}{7}$$

$$\frac{3}{2x}=\frac{1}{7}$$

$$x=\frac{21}{2}$$

x = 10.5 day
So P will do that work alone in 10.5 days

26. P takes 50% more time than Q. If they work together, the work will be done in 18 days. In how many days will Q alone complete the work?

(a) 30 days (b) 22 days
(c) 24 days (d) 25 days

RRB JE - 31/05/2019 (Shift-II)

Ans : (a) Let time taken by Q = x days
then time taken by P = 1.5x days
one day work of (P + Q)

$$\frac{1}{x}+\frac{1}{1.5x}=\frac{1}{18}$$

$$\frac{1.5+1}{1.5x}=\frac{1}{18}$$

$$\frac{2.5}{1.5x}=\frac{1}{18}$$

$$x=\frac{18\times 2.5}{1.5}=30$$

So alone Q will complete that work in 30 days.

27. Mahesh takes 18 days to complete a task alone while Kishore takes 36 days to complete the same task alone. If they work together for 6 days then what percentage of the work will remain?

(a) 50% (b) 30%
(c) 40% (d) 60%

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans. (a) : One day work of Mahesh = $\frac{1}{18}$ part

one day work of Kishore = $\frac{1}{36}$ part

one day work of both (Mahesh + Kishore) = $\frac{1}{18}+\frac{1}{36}$

$=\frac{3}{36}=\frac{1}{12}$ part

six day work of both = $\frac{6}{12}=\frac{1}{2}$ part

Remain work = $1-\frac{1}{2}=\frac{1}{2}$ part

Remain work in % = $\frac{1/2}{1}\times 100 = \frac{1}{2}\times 100=50\%$

28. Lohit, as an airman, is twice as capable as Ayush and together they finish a work in 17 days. In how many days does Ayush alone finish the same work?

(a) 34 days (b) 51 days
(c) 68 days (d) 40 days

RRB RPF SI – 13/01/2019 (Shift-II)

Ans :(b) Efficiency ratio of Lohit and Ayush = 2:1
∴ Time's ratio = 1:2
from the question,

$$\frac{1}{2x}+\frac{1}{x}=\frac{1}{17}$$

$$\frac{1+2}{2x}=\frac{1}{17}$$

$$\frac{3}{2x}=\frac{1}{17}$$

$$2x=51$$

Taken time by Ayush to complete the work = 2x = 51 days

29. X, works equal to half of Y in 1/6 time. If together they can complete a work in 10 days, then how many days will 'Y' take to do the same work alone?

(a) 24 days (b) 30 days
(c) 40 days (d) 35 days

RRB RPF Constable – 17/01/2019 (Shift-III)

Ans : (c) y completes $\frac{1}{2}$ of work in $\frac{1}{6}$ time of x

$\therefore$ y will completes the work in $\frac{1}{3}$ time of x

According to the question,

$$\frac{1}{x}+\frac{1}{y}=\frac{1}{10}, \quad \frac{1}{\frac{y}{3}}+\frac{1}{y}=\frac{1}{10}$$

$$\frac{4}{y}=\frac{1}{10}$$

y = 40 days

30. A and B together can complete a work in 35 days. If A works alone and completes $\frac{4}{7}$ part of the work and leaves the remaining work for B. Thus if the work takes 114 days to complete. So how many days will A who is more efficient in both, complete the work alone?

(a) 45 (b) 42
(c) 48 (d) 40

RRB RPF SI – 10/01/2019 (Shift-III)

Ans. (b) : Suppose A does any work in A days and B does the same work in B days

Time taken by A to do $\frac{4}{7}$ part work $=\frac{4}{7}A$

and time taken by B to do remain $\left(1-\frac{4}{7}=\frac{3}{7}\right)$ part of work

$$=\frac{3}{7}B$$

According to the question,

$$\frac{1}{A}+\frac{1}{B}=\frac{1}{35} \quad \text{----- (i)}$$

$$\frac{4}{7}A+\frac{3}{7}B=114 \quad \text{----- (ii)}$$

on solving the equation (i) and (ii)

A = 42 days

B = 210 days

So time taken to completing work by A = 42 days

31. A and B together can complete a work in 1.2 days. Although if A leaves after complete half of the work alone, then B alone completes the remaining half of the work. Thus it takes 2.5 days to complete the work. If B is 50% more efficient than A and B alone finishes the work from start to finish, then how many days will he take?

(a) 1.5 (b) 2.2
(c) 2.0 (d) 1.8

RRB RPF Constable – 19/01/2019 (Shift-III)

Ans. (c) : Let A completes any work in x day.

$\therefore$ work capacity of A $=\frac{1}{x}$

$\therefore$ work capacity of B $=\frac{1}{x}\times\frac{150}{100}=\frac{3}{2x}$

$\therefore$ According to the question,

$$\frac{1}{x}+\frac{3}{2x}=\frac{1}{1.2}$$

$$\frac{2+3}{2x}=\frac{1}{1.2}$$

$$2x=6$$

$$x=3$$

So alone B will complete that work in 2 days.

32. Kirti and Malti together can complete a task in 12 days, while Malti can finish it in 30 days. In how many days can Kirti alone complete the task?

(a) 40 days (b) 10 days
(c) 20 days (d) 30 days

RRB Group-D – 18/09/2018 (Shift-II)

Ans. (c) : One day work of (Kirti + Malti) $=\frac{1}{12}$ part

one day work of Malti $=\frac{1}{30}$ part

one day work of Kirti $=\frac{1}{12}-\frac{1}{30}=\frac{5-2}{60}=\frac{3}{60}=\frac{1}{20}$ part

$\therefore$ Time taken to complete the task by kirti = 20 days

33. Reeta and Meena together can complete a work in 10 days, while Reeta alone can complete the same work in 15 days. In how many days will Meena complete this work alone.

(a) 38 days (b) 32 days
(c) 28 days (d) 30 days

RRB Group-D – 30/10/2018 (Shift-II)

Ans : (d) Work done together in 1 day by Reeta and Meena $=\frac{1}{10}$ part

work done by Reeta in 1 day $=\frac{1}{15}$ part

work done by Meena in 1 day

$=\frac{1}{10}-\frac{1}{15}=\frac{3-2}{30}=\frac{1}{30}$ part

So Meena will complete that work in 30 day

34. Rohan and Rohit together can finish a work in 10 days. while Rohan can do the same work in 15 days alone. In how many days will Rohit alone do the same work?

(a) 32 days (b) 30 days
(c) 25 days (d) 35 days

RRB Group-D – 17/09/2018 (Shift-I)

Ans : (b) Work done in 1 day by Rohan and Rohit $= \frac{1}{10}$ part

Work done by Rohan in one day $= \frac{1}{15}$ part

Work done by Rohit in one day $= \frac{1}{10} - \frac{1}{15}$

$= \frac{3-2}{30}$

$= \frac{1}{30}$ part

So Rohit will complete that work in 30 days

35. P and Q together can finish a work in 6 days. Q alone can finish the same work in 10 days. In how many days can P alone do the same work?

(a) 15 days (b) 11 days
(c) 14 days (d) 12 days

RRB Group-D – 23/09/2018 (Shift-I)

Ans : (a) Suppose P completes this work by self in n day

1 day work of (P + Q) $= \frac{1}{6}$ part

1 day work of Q $= \frac{1}{10}$ part

1 day work of P $= \frac{1}{n}$ part

According to the question

so $\frac{1}{10} + \frac{1}{n} = \frac{1}{6}$

$\frac{1}{n} = \frac{1}{6} - \frac{1}{10}$

$\frac{1}{n} = \frac{1}{15}$

1 day work of P $= \frac{1}{15}$

So time taken to finish the work by P = 15 days

36. Working together, Sandra and Mayuri can complete a work in 45 days. However, Mayuri works alone and goes away on completion of one third of the work and then Sandra finishes the remaining work herself. As a result, both are able to complete the work in 104 days. If Mayuri work faster than her. So in how many days Sandra will be able to complete the work alone?

(a) 72 (b) 60
(c) 240 (d) 120

RRB Group-D – 28/09/2018 (Shift-III)

Ans : (d) Let sandra will complete that work in x days by self and Mayuri do that work in y days by self

one day work of both $= \frac{1}{x} + \frac{1}{y} = \frac{1}{45}$

$\therefore \quad \frac{1}{y} = \frac{1}{45} - \frac{1}{x} \Rightarrow \frac{x-45}{45x}$

$\therefore$ 1 day work of Sandra $= \frac{1}{x}$ part

$\therefore$ 1 day work of Mayuri $= \frac{x-45}{45x}$

According to the question,

$\frac{15x}{x-45} + \frac{2x}{3} = 104$

$45x + 2x^2 - 90x = 312(x-45)$

$2x^2 - 357x + 14040 = 0$

$(2x-117)(x-120) = 0$

$x = \frac{117}{2}$ or 120

$\because$ Sandra work slow so $x = 120$

37. Arjun alone can do a work in 12 days and Bheem alone can do the same work in 15 days with the help of Chetan, they together complete that work in 5 days. How many days will Chetan alone take to do that work ?

(a) 20 days (b) 24 days
(c) 15 days (d) 16 days

RRB Group-D – 28/11/2018 (Shift-I)

Ans : (a) One day work of Arjun $= \frac{1}{12}$ part

One day work of Bheem $= \frac{1}{15}$ part

Let, one day work of Chetan $= \frac{1}{x}$ part

One day work of all three $= \frac{1}{12} + \frac{1}{15} + \frac{1}{x}$

$\frac{1}{5} = \frac{5+4}{60} + \frac{1}{x}$

$\frac{1}{x} = \frac{1}{5} - \frac{9}{60}$

$\frac{12-9}{60} = \frac{1}{x}$

$\frac{1}{x} = \frac{3}{60} = \frac{1}{20}$

So time taken by Chetan to finish the work alone = 20 days

38. Tom and Jerry work together to build a wall in 4 days. Tom alone takes 6 days to complete the work. How long will it take Jerry to complete this task alone.

(a) 13 days (b) 12 days
(c) 16 days (d) 10 days

RRB Group-D – 16/11/2018 (Shift-III)

Ans. (b) : Let time taken to complete the work by Jerry = x days
According to the question,

$$\frac{1}{6}+\frac{1}{x}=\frac{1}{4}$$

$$\Rightarrow \frac{x+6}{6x}=\frac{1}{4}$$

$\Rightarrow 4x + 24 = 6x$

$\Rightarrow 2x = 24 \Rightarrow x = 12$ day

So Jerry will complete the work in 12 days alone.

39. Ishan is twice as good a worker as Kamal and both together finish a work in 29 days. In how many days will Kamal alone can do this work.

(a) 58 (b) 70
(c) 87 (d) 116

RRB NTPC 17.01.2017 Shift-1

Ans : (c) Let Kamal do any work in x days

Ishan do that work in $\frac{x}{2}$ day

According to the question,

work done by both in a day =

$$\frac{1}{x}+\frac{2}{x}=\frac{1}{29}$$

$$\frac{3}{x}=\frac{1}{29}$$

$x = 29 \times 3$

$\boxed{x=87}$ days

40. G and H can do a work in 30 days. If H alone can do it in 50 days. Then G alone can do it in _____ days.

(a) 75 (b) 70
(c) 60 (d) 65

RRB NTPC 17.01.2017 Shift-2

Ans : (a) Let G will complete that work alone in x days
According to the question,

$$\therefore \frac{1}{50}+\frac{1}{x}=\frac{1}{30}$$

$$\Rightarrow \frac{1}{x}=\frac{1}{30}-\frac{1}{50}=\frac{5-3}{150}$$

$$x=\frac{150}{2}=75 \text{ days}$$

So G will do that work alone 75 days

41. Amrit has twice capacity of painting than Kushal. Together they complete a work in 6 days. In how many days Kushal will complete that work alone?

(a) 10 (b) 12
(c) 24 (d) 18

RRB NTPC 17.01.2017 Shift-3

Ans : (d) Ability's ratio of Amrit and Kushal = 2 : 1

$\therefore$ Time's ratio = 1 : 2

Let time taken by Amrit and Kushal to finish the work is x and 2x days.
from the question-

$$\frac{x\times 2x}{x+2x}=6$$

$$\frac{2x^2}{3x}=6 \Rightarrow x=\frac{6\times 3}{2}$$

$\boxed{x=9 \text{ day}}$

So time taken to complete the work by Kaushal= 9 × 2 = 18 days

42. M can do a work in 5 days. R can do the same work in 20 days. In how many days will both of them finish the same work?

(a) 4 days (b) 3 days
(c) 2 days (d) 1 days

RRB NTPC 05.04.2016 Shift-1

Ans : (a) Time taken to complete work together by M and R

$$=\frac{5\times 20}{5+20}$$

$$=\frac{100}{25}$$

= 4 days

43. E and F can do a work in 10 days. If E alone can do it in 30 days, F alone can do it in ___ days.

(a) 15 (b) 20
(c) 25 (d) 18

RRB NTPC 05.04.2016 Shift : 3

Ans :(a) Work done by F in 1 day $=\frac{1}{10}-\frac{1}{30}$

$$=\frac{3-1}{30}=\frac{2}{30}=\frac{1}{15} \text{ part}$$

So F can do that work alone in 15 days

44. P and Q together can complete a work in 12 days. P alone does the same work in 30 days. In how much time will Q alone complete the same work?

(a) 20 days (b) 30 days
(c) 25 days (d) 35 days

RRB NTPC 04.04.2016 Shift : 3

Ans : (a) Time taken By Q to completing that work alone.

$$=\frac{12\times 30}{30-12}=\frac{12\times 30}{18}$$

= 20 days

45. Sanju takes 16 days to complete a work. If she works with her friend Jenny to complete it in 12 days. How many days will it take for jenny to complete the work alone?

(a) 48 (b) 32
(c) 36 (d) 24

RRB NTPC 03.04.2016 Shift : 2

Ans : (a) one day work of Jenny $= \frac{1}{12} - \frac{1}{16}$

$= \frac{4-3}{48} = \frac{1}{48}$ part

So Jenny will complete the work alone in 48 days

46. P is twice as capable of doing work as Q. Together they complete a work in 22 days. In how much time Q will it take to complete this work?

(a) 33 days (b) 22 days
(c) 66 days (d) 99 days

RRB NTPC 02.04.2016 Shift : 1

Ans : (c) Let P takes x days to do work and Q takes 2x days to do work.

According to the question,

$\therefore \frac{1}{x} + \frac{1}{2x} = \frac{1}{22}$

$\frac{2+1}{2x} = \frac{1}{22}$

$x = 33$

$\therefore$ Q will take 66 days to complete work.

47. Manoj is twice capable fisherman than Anurag and together they finish a work in 22 days. In how many days will Anurag alone can do the same work.

(a) 44 (b) 66
(c) 88 (d) 60

RRB NTPC 11.04.2016 Shift : 2

Ans : (b) Let Manoj will complete the work in x days. Then Anurag will complete that work in 2x days.

so $\frac{1}{x} + \frac{1}{2x} = \frac{1}{22}$

$\frac{2+1}{2x} = \frac{1}{22}$

$\frac{3}{2x} = \frac{1}{22}$

$2x = 66$

$x = 33$

So Anurag will complete that work = 2x = 66 days

48. Ullas has a double the working capacity of Tejas if they can finish a work together in 18 days, them in how many days will Tejas complete the same work alone?

(a) 36 (b) 54
(c) 29 (d) 72

RRB NTPC 11.04.2016 Shift : 3

Ans : (b) Let Ullas does work in x days and Tejas in 2x days.

According to the question,

$\frac{1}{x} + \frac{1}{2x} = \frac{1}{18}$

$\frac{2+1}{2x} = \frac{1}{18}$

2x = 54 days

49. Yash is twice as capable draftsman as Vivek and they finish a work together in 23 days. In how many days will Vivek alone complete the same work?

(a) 46 (b) 69
(c) 92 (d) 60

RRB NTPC 07.04.2016 Shift : 2

Ans : (b)

	Yash	Vivek
Ability	2x	x
Taken time	x	2x

from the question,

$\frac{1}{x} + \frac{1}{2x} = \frac{1}{23}$

$\Rightarrow \frac{3}{2x} = \frac{1}{23}$

$\Rightarrow x = \frac{69}{2}$ day

So time taken to completing the work by vivek (2x)

$= \frac{69}{2} \times 2 = 69$ days

50. Himanshu is twice as capable as Ankit as a wood cutter and together they finish a work in 16 days. In how many days will Ankit alone complete the same work?

(a) 32 (b) 48
(c) 64 (d) 40

RRB NTPC 26.04.2016 Shift : 3

Ans : (b) Let Ankit complete the work in 2x day and Himanshu in x day.

According to the question,

$\frac{1}{x} + \frac{1}{2x} = \frac{1}{16}$

$\frac{2+1}{2x} = \frac{1}{16}$

$2x = 48$

$x = 24$

So Ankit will do that work in (2x) = 24 × 2 = 48 days

51. X does 25% of a work in 20 days. Y joins up with X and they together do the remaining work in 15 days. So in how many days can Y alone do the same work?

(a) 30 days (b) $25\frac{1}{2}$ days
(c) $26\frac{2}{3}$ days (d) $26\frac{1}{3}$ days

RRB NTPC 27.04.2016 Shift : 2

Ans : (c) X complete 25% of a work $= \frac{1}{4}$ part = in 20 days

$\therefore$ X will complete whole work = 80 days

Remain work $= 1 - \frac{1}{4} = \frac{3}{4}$ part

Both completes $\frac{3}{4}$ part of work together in 15 days

$\therefore$ Both will complete whole work together $= 15 \times \frac{4}{3} = 20$ days

$\therefore$ one day work of Y $= \frac{1}{20} - \frac{1}{80} = \frac{4-1}{80} = \frac{3}{80}$

Y will complete the work $= \frac{80}{3} = 26\frac{2}{3}$ days

52. As a sailar, Anirudh is twice as fast as Ashwin and together they finish a work in 14 days. In how many days Ashwin alone can complete the work.

(a) 28 (b) 42
(c) 56 (d) 35

RRB NTPC 30.04.2016 Shift : 3

Ans : (b) Let Anirudh and Ashwin finishes the work in x days and 2x days

then one day work of Anirudh $= \frac{1}{x}$ part

and one day work of Ashwin $= \frac{1}{2x}$ part

According to the question,

then $\frac{1}{x} + \frac{1}{2x} = \frac{1}{14}$ part

$\frac{2+1}{2x} = \frac{1}{14}$

$\frac{3}{2x} = \frac{1}{14}$

$x = 21$

So time taken by Ashwin to complete the work alone

$= 2x = 2 \times 21 = 42$ days

Type - 3

53. P and Q can complete a work in 12 days and 16 days respectively. They work together for 4 days. How much of the work is left?

(a) 3/4 (b) 7/12
(c) 5/12 (d) 6/7

RRB JE - 27/05/2019 (Shift-I)

Ans : (c) One day work of P $= \frac{1}{12}$ part

one day work of Q $= \frac{1}{16}$ part

four days work of (P + Q) $= \left(\frac{1}{12} + \frac{1}{16}\right) \times 4$

$= \frac{(4+3)}{48} \times 4 = \frac{7}{48} \times 4 = \frac{7}{12}$ part

remain work $= 1 - \frac{7}{12}$

$= \boxed{\frac{5}{12} \text{ part}}$

54. Ajay and Vijay can do a work in 70 and 60 days respectively. They start working together and after some time, Ajay leaves. Vijay completed the remaining work in 47 days, when did Ajay quit the work ?

(a) After 7 days (b) After 14 days
(c) After 10 days (d) After 8 days

RRB JE - 26/06/2019 (Shift-III)

Ans : (a) Let Ajay does work till x days.

According to the question,

$\frac{x}{70} + \frac{x+47}{60} = 1$

$60x + 70(x+47) = 4200$

$60x + 70x + 3290 = 4200$

$130x = 4200 - 3290$

$130x = 910$

$x = \frac{910}{130} = 7$

So Ajay left work in after 7 days

55. A can finish a work in 30 days and B can finish the same work in 20 days. A and B do the work together for 6 days and after that A leaves the work. In how many days B will complete the remaining work?

(a) 15 days (b) 16 days
(c) 10 days (d) 18 days

RRB Group-D – 08/10/2018 (Shift-I)

Ans. (c) LCM of A and B = 60

So total work = 60 unit

work done by A in one day $= \frac{60}{30} = 2$ unit

work done by B in one day $= \frac{60}{20} = 3$ unit

one day work of both (A + B) = 2 + 3 = 5 unit

6 day both of work = 5 × 6 = 30 unit

Remain work = 60–30 = 30 unit

Time taken to do remain work by B $= \frac{30}{3} = 10$ days

56. A alone can finish a work in 3 days. B alone can finish this work in 7 days. If A and B work together for 2 days, then what part of work will be left?

(a) $\frac{1}{7}$ (b) $\frac{4}{21}$
(c) $\frac{2}{21}$ (d) $\frac{1}{21}$

RRB Group-D – 20/09/2018 (Shift-II)

Ans : (d)

one day work of A $= \frac{1}{3}$ Part

one day work of B $= \frac{1}{7}$ part

one day work of both (A + B) $=\left(\frac{1}{3}+\frac{1}{7}\right)$

$=\frac{7+3}{21}=\frac{10}{21}$ part

Two days work of (A + B) $=\frac{10\times 2}{21}=\frac{20}{21}$ part

so remain part of the work $=1-\frac{20}{21}$

$=\frac{1}{21}$ part remaining.

57. Working alone, Rakesh and Ranjit can complete a task in 4 and 6 days respectively. Ranjit started work alone and Rakesh joined it after 2 days. How many days will they need to complete the remaining work?

(a) 1.6 (b) 2
(c) 0.8 (d) 1.2

RRB Group-D – 22/09/2018 (Shift-I)

Ans : (a) Let remain work complete in x days.
∴ Rakesh will complete the work in = x days
and Ranjit will complete the work in = x + 2 days

∴ one day work of both = $\frac{x+2}{6}+\frac{x}{4}=1$

$\frac{4x+8+6x}{24}=1$

$10x = 24 - 8$

$10x = 16$

$x = 1.6$

58. 18 people can complete a work in 12 days. After 4 days of work, 2 workers left the work. In how many days will the remaining work be completed after the workers left the work?

(a) 8 (b) 6
(c) 9 (d) 10

RRB Group-D – 08/10/2018 (Shift-II)

Ans : (c) Total work= 18 × 12
work done = 18 × 4
remain work = 18 × 8
18 × 8 = (18 – 2) × D
18 × 8= 16 × D
D = 9 days

59. 800 men complete a work in 30 days, 80 men leave the work after working 12 days. In how many days can the remaining work be completed by the remaining people.

(a) 22 (b) 20
(c) 28 (d) 24

RRB Group-D – 15/10/2018 (Shift-I)

Ans : (b) Total work = 800 × 30
done work = 800 × 12
remain work = 800 × 18

According to the question,
800 × 18 = (800 - 80) × D
800 × 18 = 720 × D

D = $\frac{800\times 18}{720}=20$

60. Ranjit can complete a work in 25 days while Anji can finish it in 20 days. They work together for 5 days and then Ranjit leaves. How many days will Anji take to finish the remaining work?

(a) 10 days (b) 9 days
(c) 11 days (d) 15 days

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (c) According to the question,

One day work of Ranjit $=\frac{1}{25}$ part

One days work of Anji $=\frac{1}{20}$ part

∴ One days work of both (Ranjit + Anji) $=\frac{1}{25}+\frac{1}{20}$

$=\frac{4+5}{100}=\frac{9}{100}$ part

∴ 5 days work of (Ranjit + Anji) $=5\times\frac{9}{100}=\frac{9}{20}$ part

so Anji will complete remain work

$=\left(1-\frac{9}{20}\right)\times 20=\frac{11}{20}\times 20=11$ days

61. Sandra and Mayuri can work together and complete a task in 60 days. Although Mayuri works alone and leaves after completing one third of the work and Sandra then finishes the remaining work alone. Thus, both of them completed the work in 150 days. If Mayuri worked faster than Sandra in how many days Sandra could do that work alone?

(a) 240 (b) 180
(c) 165 (d) 225

RRB Group-D – 06/12/2018 (Shift-III)

Ans. (b)
Let time taken by sandra to completing work = x day
Let time taken by Mayuri to completing work = y day

∴ one day work of both = $\frac{1}{x}+\frac{1}{y}=\frac{1}{60}$.....(i)

Time taken by Mayuri in completing $\frac{1}{3}$ of work = $\frac{y}{3}$ days

Remain work = $1-\frac{1}{3}=\frac{2}{3}$

Time taken by Sandra in completing $\frac{2}{3}$ part of work = $x\times\frac{2}{3}=\frac{2x}{3}$ day

Time taken by both to complete the work

$= \frac{2x}{3} + \frac{y}{3} = 150$

$2x + y = 450$ .....(ii)

From equation (ii)

$y = 450 - 2x$ .....(iii)

on putting value of y in (i)

$60 (x + y) = xy$

$60 (x + 450 - 2x) = x (450 - 2x)$

$60 (450 - x) = 450x - 2x^2$

$60 \times 450 - 60x = 450x - 2x^2$

$2x^2 - 510x + 60 \times 450 = 0$

$x^2 - 255x + 13500 = 0$

$x^2 - (180 + 75) x + 13500 = 0$

$x^2 - 180x - 75x + 13500 = 0$

$x (x-180) - 75 (x-180) = 0$

$(x-180) (x-75) = 0$

$x = 180, 75$

So Sandra (Sandra works slower than Mayuri) alone will take time to complete the work = 180 days

62. 280 men complete a work in 30 days. After 4 days of work, 80 men left the work. How many days will it take to complete the remaining work?

(a) $27\frac{2}{5}$ (b) $34\frac{2}{5}$

(c) $33\frac{2}{5}$ (d) $36\frac{2}{5}$

RRB Group-D – 15/10/2018 (Shift-III)

Ans. (d) : Remain days = 30 – 4 = 26 days

remain male = 280 – 80 = 200 men

If 280 men do work till 26 days then work will finish

from $M_1D_1 = M_2 \times D_2$

$280 \times 26 = 200 \times D_2$

$D_2 = \frac{280 \times 26}{200} = \frac{182}{5}$

$= 36\frac{2}{5}$ days

63. S and T can finish a work in 50 days. They worked together for 20 days and then left. How much of the work is left?

(a) 3/5 (b) 1/3

(c) 1/2 (d) 2/5

RRB NTPC 30.03.2016 Shift : 1

Ans : (a) one day's work of S and T $= \frac{1}{50}$ part

20 days work of S and T $= \frac{20}{50} = \frac{2}{5}$ part

$\therefore$ remain work $= 1 - \frac{2}{5} = \frac{3}{5}$ part

64. X can copy 60 pages in 4 minutes, X and Y together can copy 750 pages in 30 minutes. In how many minutes can 'Y' copy 100 pages?

(a) 8 (b) 16

(c) 10 (d) 5

RRB NTPC 19.04.2016 Shift : 3

Ans : (c) 1 minute work of X $= \frac{60}{4} = 15$ page

1 minute work of X and Y $= \frac{750}{30} = 25$ page

$\therefore$ one minute work of Y = 25 – 15 = 10 page

$\therefore$ Time taken in copying to 100 pages by Y

$= \frac{100}{10} = 10$ minute

65. A completes 2/5 part of a work in X days. After then calls B and they together finish the remaining work in 6 days. If B alone takes 100/6 days. Then find the value of X.

(a) 10 days (b) 50/3 days

(c) 20 days (d) 40 days

RRB NTPC 18.04.2016 Shift : 2

Ans : (a)

X days work of A $= \frac{2}{5} \Rightarrow$ complete work $= \frac{5}{2}x$

remain work $= 1 - \frac{2}{5} = \frac{3}{5}$

time taken by A and B together to finish $\frac{3}{5}$ work = 6

time taken by A and B to finish the whole work together

$= 6 \times \frac{5}{3} = 10$ days

According to the question,

$\frac{1}{\frac{5}{2}x} + \frac{1}{\frac{100}{6}} = \frac{1}{10}$

$\frac{2}{5x} + \frac{6}{100} = \frac{1}{10} \Rightarrow \frac{40 + 6x}{100x} = \frac{1}{10}$

$40 + 6x = 10x$

$4x = 40$

$x = 10$ days

66. P and Q can complete a work in 12 days, Q and R in 15 days, R and P in 20 days. In how many days can they finish the work together?

(a) 10 (b) 12

(c) 5 (d) 14

RRB NTPC 22.04.2016 Shift : 1

Ans : (a) work done by (P + Q) in 1 day $= \frac{1}{12}$ part

work done by (Q + R) in 1 day $= \frac{1}{15}$ part

work done by (R+ P) in 1 day $= = \frac{1}{20}$ part

to adding-

work done by 2(P + Q + R) in 1 days $= \frac{1}{12} + \frac{1}{15} + \frac{1}{20}$

$= \frac{5+4+3}{60}$

$= \frac{12}{60}$

$= \frac{1}{5}$ part

$\therefore$ work done by (P + Q + R) in 1 days $= \frac{1}{5 \times 2}$

$= \frac{1}{10}$ part

So time taken by (P + Q + R) to finish the work together = 10 days

Type - 3

67. Arun and Amit can do a piece of work in 9 days and 12 days respectively. If they work for alternate days and Amit starts the work first, then in how many days 35/36 parts of whole work will be completed?

(a) 10 days (b) 12 days
(c) 5 days (d) 8 days

RRB RPF Constable – 20/01/2019 (Shift-III)

Ans : (a) 2 days work of Amit and Arun

$= \left(\frac{1}{12} + \frac{1}{9}\right) = \left(\frac{3+4}{36}\right) = \frac{7}{36}$ part

$\therefore$ 2 × 5 = 10 days work $= \frac{7 \times 5}{36}$

$= \frac{35}{36}$ part

$\therefore$ Required time = 10 days

68. P, Q and R can do a work separately in 10, 12 and 15 days respectively. They start working together and after 2 days P quits working. Q quit working 3 days before the work was completed. In how many days will the work be completed?

(a) 6 (b) 7
(c) 5 (d) 8

RRB JE - 29/05/2019 (Shift-II)

Ans : (b) Let total work completed in x days
According to the question,

$\frac{2}{10} + \frac{x-3}{12} + \frac{x}{15} = 1$

$\frac{1}{5} + \frac{x-3}{12} + \frac{x}{15} = 1$

$\frac{5(x-3)+4x}{60} = \frac{4}{5}$

9x – 15 = 48

9x = 63

x = 7 days

So work will complete in 7 days

69. A and B can complete a piece of the work in 15 days and B alone can do it in 18.75 days. They start working together but B leaves 7.5 days before the work is done. How many days did they work together?

(a) 13.75 (b) 11.25
(c) 13.5 (d) 12.5

RRB RPF SI – 06/01/2019 (Shift-III)

Ans : (c) According to the first statement.

work done by A and B in a day $= \frac{1}{15}$ part

work done by B in a day $= \frac{1}{18.75}$ part

work done by A in a day

$\frac{1}{A} = \frac{1}{15} - \frac{1}{18.75}$

$\frac{1}{A} = \frac{1}{15} - \frac{4}{75}$

$\frac{1}{A} = \frac{5-4}{75}$

$\frac{1}{A} = \frac{1}{75}$

So time taken by A to finish the work is 75 days.
Let work completes in x days
According to the second statement

$\frac{x}{75} + \frac{(x-7.5)}{18.75} = 1$

$\frac{x}{75} + \frac{x}{18.75} - \frac{7.5}{18.75} = 1$

$\frac{18.75x + 75x}{75 \times 18.75} = 1 + \frac{7.5}{18.75}$

$\frac{93.75x}{75 \times 18.75} = \frac{18.75 + 7.5}{18.75}$

$\frac{93.75x}{75} = 26.25$

$x = \frac{26.25 \times 75}{93.75} = 21$

So both worked together for = 21 – 7.5 = 13.5 days

70. A and B together can complete a work in 15 days, while A alone can complete the work in 18.75 days. They start working together but A leaves work 12.5 days before the work is completed. How long did A and B work together

(a) 10 (b) 13.75
(c) 11.25 (d) 12.5

RRB RPF SI – 12/01/2019 (Shift-III)

Ans. (d) : one day work of A + B = $\frac{1}{15}$ part

one day work of B =

$\frac{1}{15}-\frac{1}{18.75}=\frac{18.75-15}{15\times18.75}=\frac{3.75}{15\times18.75}=\frac{0.2}{15}=\frac{2}{150}=\frac{1}{75}$

one day work of B = $\frac{1}{75}$ part

12.5 days work of B = $\frac{12.5}{75}=\frac{125}{750}=\frac{1}{6}$ part

remaining work = $1-\frac{1}{6}=\frac{5}{6}$ part

time taken by (A+B) in completing $\frac{5}{6}$ work = $15\times\frac{5}{6}$

=12.5 days

So A and B worked together for 12.5 days.

71. Amod and Pramod can work separately and paint a wall in 15 and 24 days respectively. They start working together, but Amod leaves 11 days before the work is complete. How many days did it take to complete this work?

(a) 16 (b) 17
(c) 15 (d) 18

RRB Group-D – 05/10/2018 (Shift-II)

Ans. : (a) Let work will be complete in x days.
According to the question,

$\frac{x-11}{15}+\frac{x}{24}=1$

24x – 264 + 15x = 360

39x = 624

$x=\frac{624}{39}=16$

So total work (x) will be complete in 16 days.

72. 15 person took up the task of digging a pond in 20 days. After 10 days, 5 people left the work in the next 5 days 5 workers also left the work. How long will it take to complete the work?

(a) 45 (b) 25
(c) 55 (d) 35

RRB Group-D – 10/10/2018 (Shift-II)

Ans : (d) M.D = $M_1D_1 + M_2D_2 + M_3D_3$

$15 \times 20 = 15 \times 10 + 5 \times 10 + 5 \times D_3$

$300 - 150 - 50 = 5D_3$

$5D_3 = 100$

$D_3 = 20$

Total time taken = 20 + 10 + 5 = 35 days

73. Pavitra can do a work in 8 days, Dinu can do the same work in 10 days and Naba can do it in 12 days. They started working together, but Pavitra left after two days and the remaining work was done by Dinu and Naba. How long did it take to complete the whole work.

(a) 67/11 days (b) 45/11 days
(c) 23/11 days (d) 50/11 days

RRB Group-D – 29/10/2018 (Shift-III)

Ans : (b) Let work will be complete in x days
According to the question,

$\frac{2}{8}+\frac{x}{10}+\frac{x}{12}=1$

$\frac{1}{4}+\frac{x}{10}+\frac{x}{12}=1$

$\frac{x}{10}+\frac{x}{12}=1-\frac{1}{4}$

$\frac{6x+5x}{60}=\frac{3}{4}$

$\frac{11x}{60}=\frac{3}{4}$

$x=\frac{45}{11}$ days

74. 48 people can do a work in 17 days. After 6 days 4 workers left the work. After that how many days will it take to complete the work?

(a) 16 (b) 15
(c) 12 (d) 13

RRB Group-D – 17/09/2018 (Shift-II)

Ans : (c) Let remaining work completes in x days
from $M_1D_1 = M_2D_2$

$48 \times (17 - 6) = 44 \times x$

$48 \times 11 = 44 \times x$

$x=\frac{48\times11}{44}=12$

75. A and B can do a work in 45 hours. If A works alone and completes the 3/8 part of the work, then leaves the remaining work to be done by B. It takes a total of 102 days to complete the work. In how many days A will complete the work, which is more efficient between them to complete the work alone?

(a) 170 (b) 96
(c) 72 (d) 120

RRB Group-D – 05/12/2018 (Shift-II)

Ans. (c) Let A completes total work in x days

time taken by A in complete $\frac{3}{8}$ part = $\frac{3x}{8}$ days

time taken by B in complete $\frac{5}{8}$ part = $\left(102-\frac{3x}{8}\right)$ day

time taken by B to completing whole work $=\left(102-\frac{3x}{8}\right)\frac{8}{5}$ day

According to the question,

$\frac{1}{x}+\frac{1}{\left(102-\frac{3x}{8}\right)\frac{8}{5}}=\frac{1}{45}$

$\frac{1}{x}+\frac{5}{816-3x}=\frac{1}{45}$

$\frac{816-3x+5x}{(816-3x)x}=\frac{1}{45}$

$15(816 + 2x) = x(272 - x)$
$12240 + 30x = 272x - x^2$
$x^2 - 272x + 30x + 12240 = 0$
$x^2 - 242x + 12240 = 0$
$x^2 - 72x - 170x + 12240 = 0$
$(x - 72)(x - 170) = 0$
$x = 72, 170$
So, A will complete the whole work in 72 days.

76. 36 men can do a work in 24 days. 4 workers left work after 8 days. Since then how many days will it take to complete the work?
(a) 18 (b) 16
(c) 12 (d) 20
RRB Group-D – 05/12/2018 (Shift-II)

Ans. (a) After 8 days-
remaining men = 32
remaining days = 16
Later, the remaining 16 days of work for 36 men is left, which will be complete by 32 men together.
$\because M_1D_1 = M_2D_2$
$\therefore 36 \times 16 = 32 \times D_2$
$D_2 = 18$ days

77. A and B complete a work in 12 and 16 days respectively. Both of them work together for 3 days then A leaves. In how many days only B will complete the remaining work?
(a) 12 (b) 18
(c) 9 (d) 10
RRB Group-D – 05/11/2018 (Shift-I)

Ans. (c) :
Time taken by A to completing the work = 12 days
Time taken by B to completing same work = 16 days
1 day work of A and B $= \frac{1}{12} + \frac{1}{16} = \frac{4+3}{48} = \frac{7}{48}$ part
$\therefore$ 3 day work $= 3 \times \frac{7}{48} = \frac{7}{16}$ part
remaining work $= 1 - \frac{7}{16} = \frac{9}{16}$ part
$\because$ Time taken by B to completing $\frac{1}{16}$ part of work · 1 day
$\therefore$ Time taken by B to completing $\frac{9}{16}$ part of work
$= 16 \times \frac{9}{16} = 9$ days
So time taken by B to completing remaining work = 9 days

78. A and B can complete a work in 12 days. Although alone A completed 1/5 of the work and left the work. Then B started working alone and completed the remaining work. The work take a total of 22 days to complete. If B is more efficient than A, how many days did B take to complete the work alone?
(a) 16.5 (b) 16.5 or 20
(c) 30 or 44 (d) 20
RRB Group-D – 12/11/2018 (Shift-II)

Ans : (b) $\frac{1}{A} + \frac{1}{B} = \frac{1}{12}$ ----(i)
Suppose A takes x days to doing $\frac{1}{5}$ part of work.
time taken in completing whole work = 5x days
time taken by B to doing $\frac{4}{5}$ part of work = (22 – x) day
time taken in completing whole work = $(22 - x)\frac{5}{4}$ day
from the equation (i)
$\Rightarrow \frac{1}{5x} + \frac{4}{5(22-x)} = \frac{1}{12}$
$\Rightarrow \frac{110 - 5x + 20x}{25x(22-x)} = \frac{1}{12}$
$\Rightarrow \frac{110 + 15x}{25x(22-x)} = \frac{1}{12}$
$\Rightarrow \frac{(22+3x)}{5x(22-x)} = \frac{1}{12}$
$\Rightarrow 264 + 36x = 110x - 5x^2$
$\Rightarrow 5x^2 - 74x + 264 = 0$
$\Rightarrow 5x^2 - 30x - 44x + 264 = 0$
$\Rightarrow 5x(x-6) - 44(x-6) = 0$
$\Rightarrow (x-6)(5x-44) = 0$
$x = 6, x = \frac{44}{5} = 8.8$
time taken by B to doing whole work $= (22 - x) \times \frac{5}{4}$ day
$= (22 - 6) \times \frac{5}{4}$
$= 16 \times \frac{5}{4} = 20$ day
or
$= (22 - 8.8) \times \frac{5}{4}$
$13.2 \times \frac{5}{4} = 16.5$ days

79. A can do a work in 8 days, while B takes 10 days to complete the same work. They started working together but A left the work 1 day before the work completed. Then how many days did A is work?
(a) 3.5 (b) 4
(c) 2 (d) 3
RRB Group-D – 23/10/2018 (Shift-II)

Ans. (b) : Work done by A in 1 days = $\frac{1}{8}$ part
work done by B in 1 day = $\frac{1}{10}$ part

work done by (A + B) in 1 day = $\frac{1}{8}+\frac{1}{10}=\frac{9}{40}$ part

Let both A and B does work together in x days.

According to the questions,

$\frac{9x}{40}+\frac{1}{10}=1$

$\Rightarrow \frac{9x}{40}=1-\frac{1}{10}$

$\Rightarrow x=\frac{9}{10}\times\frac{40}{9}$

$x=4$ days

so A worked for 4 days.

80. 28 days of food items are available for 1200 soldiers posted in a fort. After 4 days some soldiers left the fort, due to which the available food item use for 32 days. How many soldiers left the fort?

(a) 300 (b) 280

(c) 320 (d) 290

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (a) Number of posted soldiers = 1200

Let x soldiers have lefted after 4 days

we know that $M_1 \times D_1 = M_2 \times D_2$

$1200 \times 28 = 1200 \times 4 + (1200 - x) \times 32$

$1200 \times 24 = (1200 - x) \times 32$

$28800 = 38400 - 32x$

$32x = 38400 - 28800$

$32x = 9600$

$x = 300$

81. Sunita embroiders a saree in 15 days while her sister Neha can complete the same work in 10 days. They started working together, but Neha leaves work after two days. After that how many days will Sunita take to complete the embroidery alone?

(a) 10 days (b) 20 days

(c) 16 days (d) 12 days

RRB Group-D – 09/10/2018 (Shift-II)

Ans. (a)

Let time taken to complete whole work in x days.

According to the questions,

$\frac{x}{15}+\frac{2}{10}=1$

$\frac{x}{15}=1-\frac{2}{10}$

$\Rightarrow \frac{x}{15}=\frac{8}{10}$

$\Rightarrow x=15\times\frac{4}{5}$

$x=12$

$\therefore$ Whole work will finish in 12 days.

So Sunita will take 10 days to complete the remaining work

82. In a camp 180 students have food for 20 days. If the food is to lasts for 25 days, then how many students should leave the camp?

(a) 36 (b) 24

(c) 28 (d) 40

RRB NTPC 29.03.2016 Shift : 1

Ans : (a) Let the number of students leaving of the camp = x

from the questions,

$180\times 20=(180-x)\times 25$

$180-x=\frac{180\times 20}{25}$

$180-x=\frac{3600}{25}$

$180-x=144$

$x=180-144=36$

So 36 students will have to leave the camp.

83. Smita can finish a work in 12 days and Sam can finish the same work in 9 days. Both leave the work after working together for 4 days. How much work is left?

(a) 1/2 (b) 7/9

(c) 2/9 (d) 1/4

RRB NTPC 29.03.2016 Shift : 1

Ans : (c) one day work of Smita $=\frac{1}{12}$ part

one day work of sam $=\frac{1}{9}$ part

$\therefore$ 4 days work of both $=4\left(\frac{1}{12}+\frac{1}{9}\right)$

$=4\left(\frac{3+4}{36}\right)=\frac{7}{9}$ part

remaining work $=1-\frac{7}{9}=\frac{2}{9}$ part

84. Vikram and Vivek can finish a work in 50 days. They worked together for 20 days and after that both are left the work. How much work has been done by them?

(a) 3/5 (b) 1/3

(c) 1/2 (d) 2/5

RRB NTPC 29.03.2016 Shift : 3

Ans : (d) Work done by both in 20 days $=\frac{20}{50}=\frac{2}{5}$ part

85. 30 men can do a work in 20 days, After 6 days how many people should leave this job so that the entire work is done in 26 days?

(a) 9 (b) 12

(c) 8 (d) 7

RRB NTPC 19.04.2016 Shift : 1

Ans : (a) Let the number of lefted peoples is x

According to the questions,

$30 \times 20 = 6 \times 30 + (26-6) \times (30-x)$

$600 = 180 + 20\ (30-x)$

$600 - 180 = 20(30-x)$

$(30-x) = \frac{420}{20}$

$(30-x) = 21$

$x = 9$

then the number of people left the work = 9

86. Anil can do a work in 14 days and Rohit can do the same work in 21 days. Both together work for some days and then Anil quits the work. If Rohit has worked alone for three days then the number of days taken to complete the work is.

(a) 31/5 (b) 51/5
(c) 2/15 (d) 13/5

RRB NTPC 16.04.2016 Shift : 3

Ans : (b) Let Anil worked for N days

work done by Anil in 1 day $= \frac{1}{14}$ part

work done by Rohit in 1 day $= \frac{1}{21}$ part

According to the questions,

$\frac{N}{14} + \frac{N+3}{21} = 1$

$\frac{3N+2N+6}{42} = 1 \Rightarrow 5N+6 = 42$

$5N = 42 - 6 = 36$

$N = \frac{36}{5}$

So number of days to finish work

$= \frac{36}{5} + 3 = \frac{51}{5}$

87. A cleaning company employs 42 sweepers to clean a building in 25 days. 10 days later, 12 sweepers left the job. If the cleaning work is finished in 10 days then how many more cleaning workers need to be hired?

(a) 30 (b) 32
(c) 33 (d) 21

RRB NTPC 28.04.2016 Shift : 1

Ans : (c) Suppose the number of extra employees = x

According to the questions,

$M_1 \times D_1 = M_2 \times D_2$

$(25-10) \times 42 = (30+x) \times 10$

$15 \times 42 = 10 \times (30+x)$

$30 + x = \frac{630}{10} \Rightarrow 63$

$x = 63 - 30$

$\boxed{x = 33}$

88. A and B can complete a work together in 35 days. If A works alone and completes $\frac{5}{7}$ of the work and then leaves the remaining for B, it will take a total of 90 days to complete the work. How many days would take A, who is more efficient among them to complete the entire work?

(a) 40 (b) 45
(c) 48 (d) 42

Ans : (d) Let A completes the entire work in x days

person	work	time
A	1	x day
A	$\frac{5}{7}$	$\frac{5}{7}x$ day
B	$\frac{2}{7}$	$\left(90 - \frac{5}{7}x\right) = \frac{(630-5x)}{7}$
B	1	$\left(\frac{630-5x}{7}\right) \times \frac{7}{2} = \left(\frac{630-5x}{2}\right)$

According to the questions,

$\because \quad \frac{1}{A} + \frac{1}{B} = \frac{1}{35}$

$\frac{1}{x} + \frac{2}{(630-5x)} = \frac{1}{35}$

$35(630-5x+2x) = x(630-5x)$

$7(630-3x) = (126x - x^2)$

$4410 - 21x = 126x - x^2$

$x^2 - 147x + 4410 = 0$

$x^2 - 105x - 42x + 4410 = 0$

$(x-105)(x-42) = 0$

$x = 42,\ x \neq 105$ (from options)

So A will take 42 days to complete the work.

89. A and B can complete a work together in 12 days while A alone can do it in 15 days. They start working together but A leaves 10 days before the completion of the work. For how many days did A and B work together?

(a) 9 (b) 11
(c) 8 (d) 10

RRB ALP & Tec. (29-08-18 Shift-III)

Ans : (d) 1 day work of A + B $= \frac{1}{12}$ part

1 day work of A $= \frac{1}{15}$ part

1 day work of B $= \frac{1}{12} - \frac{1}{15}$

$= \frac{5-4}{60} = \frac{1}{60}$ part

According to the questions, Let A and B did work together for x days.

x.A+x.B+10B = 1

x (A+B) + 10B = 1

$x\left(\frac{1}{12}\right) + \frac{10}{60} = 1$

$\frac{x}{12} = 1 - \frac{1}{6}$

$\frac{x}{12} = \frac{5}{6}$

x = 10 days

90. 10 people could complete a work in 8 days. After 4 days, 2 of the workers left. How many days to take compete the task?

(a) 7 (b) 8
(c) 6 (d) 5

RRB ALP & Tec. (20-08-18 Shift-III)

Ans : (d) 10 men will do total work = 10 × 8 = 80 unit
remaining work after 4 days = 80 – 40 = 40 unit
8 men will do remaining work $= \frac{40}{8} = 5$ days

91. A and B together can complete a work in 12 days. However, if A works alone, completes half work and leaves then B works alone and completes the remaining work, it takes 25 days to complete the work. If B is more efficient than A, how many days would it have taken B to do the work alone?

(a) 18 (b) 22
(c) 20 (d) 15

RRB ALP & Tec. (13-08-18 Shift-I)

Ans : (c) Let A and B does the work in x and y days.

then, $\frac{1}{x}+\frac{1}{y}=\frac{1}{12}$

$\frac{y+x}{xy}=\frac{1}{12}$

$xy=12(x+y)$ ..... (i)

According to the question-

$\frac{x}{2}+\frac{y}{2}=25$

$x+y=50$ ..... (ii)

from the equation (i) and (ii)

$xy=12\times 50$

$xy=600$ ..... (iii)

on taking x = 30 and y = 20 third eqaution will be satisfied.
so taken time by B = 20 days

Type - 3

92. P and Q can do a diffrent work in 6 and 8 days respectively with the help of R they complete the work in 3 days. If total wages is Rs. 3200 then what is the amount given to R?

(a) Rs. 320 (b) Rs. 1200
(c) Rs. 400 (d) Rs. 375

RRB JE - 23/05/2019 (Shift-III)

Ans : (c) one day work of P = $\frac{1}{6}$ part

one day work of Q = $\frac{1}{8}$ part

According to the equation,

$\frac{1}{P}+\frac{1}{Q}+\frac{1}{R}=\frac{1}{3}$

$\Rightarrow \frac{1}{6}+\frac{1}{8}+\frac{1}{R}=\frac{1}{3}$

$\Rightarrow \frac{1}{R}=\frac{1}{3}-\left(\frac{7}{24}\right)$

$\frac{1}{R}=\frac{1}{24}$

work ability ratio of P, Q and R = $\frac{1}{6}:\frac{1}{8}:\frac{1}{24}=\frac{4:3:1}{24}$

P : Q : R = 4 : 3 : 1

wages of R = $3200\times\frac{1}{8}$ = Rs. 400

93. A and B can complete a work in 50 days, B and C can complete it in 37.5 days while C and A together can complete the same work in 30 days. In how many days can each of A, B and C individually complete the same work?

(a) 50, 150 and 75 (b) 40, 60 and 120
(c) 60, 120 and 40 (d) 75, 150 and 50

RRB Group-D – 26/09/2018 (Shift-II)

Ans. (d) : one day work of (A + B)= $\frac{1}{50}$ part

one day work of (B + C) = $\frac{1}{37.5}$ part

one day work of (C + A) = $\frac{1}{30}$ part

one day work of 2(A + B + C) = $\left(\frac{1}{50}+\frac{10}{375}+\frac{1}{30}\right)$ part

2(A+B+C) = $\left(\frac{1}{50}+\frac{2}{75}+\frac{1}{30}\right)$ part

2(A+B+C) = $\left(\frac{6+8+10}{300}\right)$

A+B+C= $\frac{12}{300}=\frac{1}{25}$ part

$\frac{1}{50}+C=\frac{1}{25}$

C = $\frac{1}{25}-\frac{1}{50}=\frac{2-1}{50}=\frac{1}{50}$ part

So C could completed the work in 50 days

$A+\frac{2}{75}=\frac{1}{25}$

$A=\frac{1}{25}-\frac{2}{75}=\frac{3-2}{75}=\frac{1}{75}$ part

So it will take 75 days to complete the work by A

$\frac{1}{30}+B=\frac{1}{25}$

$B=\frac{1}{25}-\frac{1}{30}=\frac{6-5}{150}=\frac{1}{150}$ part

So work could completed by A in 75 days, B in 150 days and C in 50 days

94. A and B can complete a work in 20 days, B and C can complete it in 30 days while C and A can do the same work together in 24 days. How many days will each of B and C take to complete the work individually?

(a) 48 and 80 (b) 56 and 65
(c) 54 and 72 (d) 50 and 54

RRB ALP & Tec. (20-08-18 Shift-II)

Ans : (a) 1 day work of (A + B) $= \frac{1}{20}$ part

1 day of (B + C) $= \frac{1}{30}$ part

1 day of (C + A) $= \frac{1}{24}$ part

1 day work of the A, B, C $= \left(\frac{1}{20}+\frac{1}{30}+\frac{1}{24}\right)\times\frac{1}{2}$

$= \frac{6+4+5}{120\times 2} = \frac{1}{8\times 2} = \frac{1}{16}$

time taken to completing work by B $= \frac{1}{16}-\frac{1}{24}$

$= \frac{3-2}{48} = \frac{1}{48}$

B will complete work by self in 48 days.

taken time by C to do work by self $= \frac{1}{16}-\frac{1}{20}$

$= \frac{5-4}{80} = \frac{1}{80}$

so C will complete work alone in 80 days.
so B and C will complete their work in 48 and 80 days.

Type - 6

95. If P and Q together can complete a work in 15 days, Q and R together can do the same work in 12 days and P and R together can do it in 20 days, then how many days can the three together complete the same work.

(a) 10 days (b) 14 days
(c) 4 days (d) 6 days

RRB JE - 26/05/2019 (Shift-I)

Ans : (a)

one day work of (P + Q) $= \frac{1}{15}$ part

one day work of (Q + R) $= \frac{1}{12}$ part

one day work of (P + R) $= \frac{1}{20}$ part

work done by 2 (P + Q + R) in 1 day

$= \frac{1}{15}+\frac{1}{12}+\frac{1}{20}$

$= \frac{4+5+3}{60\times 2} = \frac{12}{60\times 2} = \frac{1}{5\times 2} = \frac{1}{10}$

so time taken to finishing the work by (P + Q + R) all three = 10 days

96. P and Q can complete a work in 8 days. Q and R can complete the same work in 12 days. P, Q and R together can complete the same work in 6 days. In how many days can P and R together can do that work?

(a) 6 days (b) 8 days
(c) 5 days (d) 4 days

RRB JE - 26/05/2019 (Shift-II)

Ans : (b)

work done by (P + Q) in 1 days $= \frac{1}{8}$ part

work done by (Q + R) in 1 days $= \frac{1}{12}$ part

work done by (P + Q + R) is 1 days $= \frac{1}{6}$ part

work done by R in a day $= \frac{1}{6}-\frac{1}{8}$

$= \frac{4-3}{24} = \frac{1}{24}$

work done by P in a one day $= \frac{1}{6}-\frac{1}{12}$

$= \frac{4-2}{24} = \frac{2}{24} = \frac{1}{12}$

total work done by (P + R) $= \frac{1}{24}+\frac{1}{12}$

$= \frac{1+2}{24} = \frac{3}{24} = \frac{1}{8}$ day

97. P and Q together can complete a work in 30 days. Q and R together can do the same work in 24 days. R and P together will take 20 days to complete the same work. How many days will it take all of them together to complete the same work?

(a) 25 (b) 8
(c) 16 (d) 15

RRB JE - 24/05/2019 (Shift-III)

Ans : (c) Work done by (P + Q) in one day $= \frac{1}{30}$ part

work done by (Q + R) in one day $= \frac{1}{24}$ part

work done by (R + P) in one day $= \frac{1}{20}$ part

$2(P+Q+R) = \frac{1}{30}+\frac{1}{24}+\frac{1}{20}$

$2(P+Q+R) = \frac{4+5+6}{120} = \frac{15}{120}$

work done by (P + Q + R) in one day $= \frac{1}{16}$ part

time taken by (P + Q + R) to complete the work = 16 days

98. A and B together can complete a work in 10 days B and C together in 12 days and C and A together in 15 days. How much time will A take to complete this work alone?

(a) 34 days (b) 24 days
(c) 20 days (d) 30 days

RRB RPF Constable – 18/01/2019 (Shift-I)

Ans : (b) one day work of (A + B) = $\frac{1}{10}$ part

one day work of (B + C) = $\frac{1}{12}$ part

one day work of (C + A) = $\frac{1}{15}$ part

one day work of 2(A + B + C) = $\frac{1}{10}+\frac{1}{12}+\frac{1}{15}$

$=\frac{6+5+4}{60}=\frac{15}{60}$

one day work of (A + B + C) = $\frac{1}{4\times 2}=\frac{1}{8}$

So one day work of A = $\frac{1}{8}-\frac{1}{12}=\frac{3-2}{24}=\frac{1}{24}$

A will finish all work alone in 24 days

99. A and B can complete a task in 40 days, B and C can complete in 30 days while C and A can complete the same task together in 24 days. How many days will each of A, B and C take to complete the task individually?

(a) 48, 96 and 32 (b) 32, 48 and 96
(c) 60, 120 and 40 (d) 40, 120 and 60

RRB ALP & Tec. (29-08-18 Shift-I)

Ans : (c) one day work of $(A+B)=\frac{1}{40}$ part

one day work of $(B+C)=\frac{1}{30}$ part

one day work of $(C+A)=\frac{1}{24}$ part

one day work of all three $2(A+B+C)=\frac{1}{40}+\frac{1}{30}+\frac{1}{24}$

$=\left(\frac{3+4+5}{120}\right)\frac{1}{2}=\frac{1}{10}\times\frac{1}{2}=\frac{1}{20}$ part

one day work of A in alone $=\frac{1}{20}-\frac{1}{30}=\frac{3-2}{60}=\frac{1}{60}$ part

So A will complete work in 60 days

one day work of B in alone $=\frac{1}{20}-\frac{1}{24}=\frac{6-5}{120}=\frac{1}{120}$ part

So B will complete work in 120 days

one day work of C in alone $=\frac{1}{20}-\frac{1}{40}=\frac{2-1}{40}=\frac{1}{40}$ part

So C will complete work in 40 days

100. Ravi, Rohan and Rajesh alone can complete a work in 10, 12 and 15 days respectively. In how many days can the work be completed, if all three work together?

(a) 4 days (b) 5 days
(c) 3 days (d) 8 days

RRB NTPC 27.04.2016 Shift : 3

Ans : (a) one day work of Ravi, Rohan and Rejesh of the three

$=\frac{1}{10}+\frac{1}{12}+\frac{1}{15}$

$=\frac{6+5+4}{60}=\frac{15}{60}$

$=\frac{1}{4}$ part

So it takes 4 days to complete the work by three people.

101. A, B and C can complete a work in 81 days. A and B together can complete the same work in 97.2 days. B and C together can complete the same work in 162 days. In how many days can B complete that work alone?

(a) 243 (b) 234
(c) 261 (d) 225

RRB Group-D – 27/09/2018 (Shift-I)

Ans. (a) one day work of A, B and C $=\frac{1}{81}$ part

one day work of A and B $=\frac{1}{97.2}$ part

one day work of B and C $=\frac{1}{162}$ part

According to the question,

$\frac{1}{A}+\frac{1}{B}+\frac{1}{C}=\frac{1}{81}$ .....(i)

$\frac{1}{A}+\frac{1}{B}=\frac{10}{972}$ .....(ii)

$\frac{1}{B}+\frac{1}{C}=\frac{1}{162}$ .....(iii)

from the equation (i) and (ii)

$\frac{1}{C}=\frac{1}{81}-\frac{10}{972}$

$\frac{1}{C}=\frac{972-810}{81\times 972}=\frac{162}{81\times 972}=\frac{1}{486}$

from equation (iii)

$\frac{1}{B}+\frac{1}{486}=\frac{1}{162}$

$\frac{1}{B}=\frac{1}{162}-\frac{1}{486}=\frac{3-1}{486}$

$\frac{1}{B}=\frac{2}{486}=\frac{1}{243}$

B = 243

So B will complete work by alone in 243 days.

102. A can do a work alone in 11 days while B takes 16.5 days to do it alone. It takes 5.5 days for them to complete the work together with C. If 'C' and 'D' work together then they can complete it in 22 days. Then in how many days can D alone finish the work?

(a) 44 (b) 55
(c) 77 (d) 66

RRB Group-D – 06/12/2018 (Shift-II)

Ans. (d) : one day work of A = $\frac{1}{11}$ part

one day work of B = $\frac{1}{16.5}$ part

one day work of (A+B+C) = $\frac{1}{11}+\frac{1}{16.5}+\frac{1}{C}=\frac{1}{5.5}$ part

one day work of C = $\frac{1}{5.5}-\left(\frac{1}{11}+\frac{1}{16.5}\right)$

$=\frac{1}{5.5}-\left(\frac{2.5}{16.5}\right)$

$=\frac{3-2.5}{16.5}=\frac{0.5}{16.5}=\frac{1}{33}$ part

$\because$ one day work of (C + D) = $\frac{1}{22}$ part

$\therefore$ one day work of D = $\frac{1}{22}-\frac{1}{33}=\frac{3-2}{66}=\frac{1}{66}$ part

So D will complete work in 66 days

103. A and B can do a work in 40 days, B and C can do the same work in 56 days, while C and A together can do the same work in 70 days. How many days will C take to complete the work alone?

(a) 210 (b) 175
(c) 245 (d) 280

RRB Group-D – 05/12/2018 (Shift-I)

Ans : (d)

one day work of A and B = $\frac{1}{40}$ part ..... (1)

one day work of B and C = $\frac{1}{56}$ part .....(2)

one day work of C and A = $\frac{1}{70}$ part .....(3)

to adding equation (1), (2) and (3)

$$2(A+B+C)=\frac{1}{40}+\frac{1}{56}+\frac{1}{70}=\frac{7+5+4}{280}=\frac{16}{280}$$

$$(A+B+C)=\frac{16}{2\times 280}=\frac{8}{280}=\frac{1}{35}.....(4)$$

from equation (4) and equation (1)

one day work of C = $\frac{1}{35}-\frac{1}{40}=\frac{8-7}{280}=\frac{1}{280}$ part

So time taken by C to completing work = 280 days

104. A takes twice as long as B and C together to do a work, C takes three times the time A and B take together. If A, B and C working together take 6 days to complete the work, in how many days will A alone complete the same work?

(a) 20 days (b) 18 days
(c) 24 days (d) 15 days

RRB Group-D – 02/11/2018 (Shift-I)

Ans. (b) According to the question,

work done by B and C = work done by 2 × A

B + C = 2A ...(i)

work done by A and B = work done by 3 × C

A + B = 3C ...(ii)

$\because$ one day work done by A, B and C

$A + B + C = \frac{1}{6}$

putting the value of B + C from equation (i)

$A + 2A = \frac{1}{6}$

$3A = \frac{1}{6} \Rightarrow \boxed{A = \frac{1}{18}}$

So A will complete this work alone in 18 days

105. Rajan can complete a certain work in 6 days. Bhavesh takes 8 days to complete the same work. Charan takes the same times as Rajan and Bhavesh to work together. How long will it take for Bhavesh and Charan to complete the work together?

(a) $2\frac{2}{5}$ days (b) $2\frac{3}{5}$ days
(c) $2\frac{7}{5}$ days (d) $2\frac{4}{5}$ days

RRB Group-D – 15/10/2018 (Shift-II)

Ans : (a) According to the question,

Rajan's work = 6 day

Bhavesh's work = 8 day

one day work of both (Rajan + Bhavesh) = $\frac{1}{6}+\frac{1}{8}$

$=\frac{4+3}{24}=\frac{7}{24}$

According to the question,

Charan works alone in the same time as Rajan and Bhavesh take to work together

one day work of Bhavesh and Charan = $\frac{1}{8}+\frac{7}{24}$

$=\frac{3+7}{24}=\frac{10}{24}$

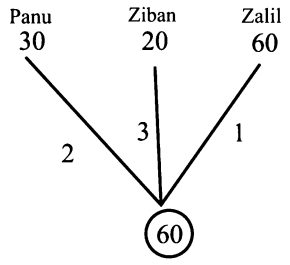
So time taken by Bhavesh and Charan to completing work = $\frac{24}{10}=2\frac{2}{5}$ day

106. Panu, Ziban and Zalil separately can finish a work in 30, 20 and 60 days respectively. Ziban started working and Panu and Zalil agreed to assist Panu on every third day. How long will it take to complete that work?

(a) 14 (b) 13
(c) 15 (d) 12

RRB Group-D – 11/10/2018 (Shift-I)

Ans : (c)



Total work = 60 unit

∵ Ziban does 3 unit work in a one day

Ziban will do in 2 day = 3 × 2 = 6 unit

∴ Panu and Zalil assist Ziban on the third day

∵ work of third day = (Ziban's work in 2 day) + (Panu + Zalil + Ziban)

= 2 × 3 + 2 + 1 + 3 = 12 unit

∵Time taken in doing 1 unit $= \frac{3}{12}$

∴ time taken in doing 60 unit $= \frac{3}{12} \times 60 =$ 15 days

107. If X and Y can complete a piece of work in 20 days, Y and Z can complete the same work in 12 days, while Z and X could do the same work in 15 days. How many days would be needed to complete the work by X, Y and Z together?

(a) 5 (b) 10
(c) 8 (d) 12

RRB Group-D – 08/10/2018 (Shift-III)

Ans : (b)

one day work of (x + y) = $\frac{1}{20}$ part

one day work of (y + z) = $\frac{1}{12}$ part

one day work of (z + x) = $\frac{1}{15}$ part

one day work of 2(x + y + z) = $\frac{1}{20}+\frac{1}{12}+\frac{1}{15}$

one day work of (x + y + z) $= \frac{3+5+4}{2\times 60}$

one day work of (x + y + z) $= \frac{12}{2\times 60} = \frac{1}{10}$ part

So the three of x, y and z will complete that work in 10 days

108. A can do a work in 6 days. B takes 8 days to complete the work. C takes equal time to the time taken by A and B to work together to complete the work. If B and C work together how much time to take them to complete the work?

(a) $\frac{14}{5}$ days (b) $\frac{13}{5}$ days
(c) $\frac{11}{5}$ days (d) $\frac{12}{5}$ days

RRB Group-D – 01/10/2018 (Shift-II)

Ans. (d) : work done by A in a day = $\frac{1}{6}$ part

work done by B in a day = $\frac{1}{8}$ part

work done by C in a day = $\frac{1}{6}+\frac{1}{8}$ part

$= \frac{4+3}{24} = \frac{7}{24}$ part

one day work with B and C together

$= \frac{1}{8}+\frac{7}{24} = \frac{10}{24} = \frac{5}{12}$ part

So, time taken by B and C to completing work with together

$= \frac{12}{5}$ day

Type - 3

109. P, Q, R and S can complete a work in 20 days, If P and Q together can complete it in 50 days and R alone can complete it in 60 days then in how many days can S can complete that work alone?

(a) 65 (b) 60
(c) 90 (d) 75

RRB JE - 24/05/2019 (Shift-III)

Ans : (d) 1 day work of $P+Q+R+S = \frac{1}{20}$ part

1 day work of $(P+Q) = \frac{1}{50}$ part

1 day work of $R = \frac{1}{60}$ part

1 day work of $S = \frac{1}{20}-\frac{1}{50}-\frac{1}{60}$

$= \frac{15-6-5}{300} = \frac{4}{300}$

So S will do this work alone in 75 days.

110. P, Q and R together complete a task. P and Q finish 70% of the work while Q and R finish 50% of the work. If they work alone, who will do the work first?

(a) R
(b) P
(c) can not determine
(d) Q

RRB JE - 26/05/2019 (Shift-III)

Ans : (b) If P, Q and R complete some work in 1 day

If 1 day work of P and Q $= 70\% = \frac{7}{10}$

1 day work of Q and R $= 50\% = \frac{1}{2}$

$\therefore$ one day work of R $= 1 - \frac{7}{10} = \frac{3}{10}$

So, R will complete work alone in $= \frac{10}{3} = 3\frac{1}{3}$ days

again, one day work's of P $= 1 - \frac{1}{2} = \frac{1}{2}$

So P will complete work alone in 2 days.

again 1 day work of Q $= \frac{1}{2} - \frac{3}{10} = \frac{2}{10} = \frac{1}{5}$

So Q will complete work alone in 5 days

At first time P will complete work.

111. A, B and C can complete a work in 81 days. A and B together can complete the same work in 97.2 days. B and C together can complete the same work in 162 days. In how many days can B complete the work alone?

(a) 225 (b) 234
(c) 243 (d) 261

RRB RPF SI – 10/01/2019 (Shift-III)

Ans : (c) According to the question,

Work done by A, B and C in 1 day $= \frac{1}{81}$

$$A + B + C = \frac{1}{81} \quad \text{..... (i)}$$

then, 1 day work of A + B $= \frac{1}{97.2}$

$$A + B = \frac{10}{972}$$

$$A + B = \frac{5}{486} \quad \text{..... (ii)}$$

on putting the value of equation (ii) into equation (i)

$$C = \frac{1}{81} - \frac{5}{486}$$

$$C = \frac{6-5}{486} = \frac{1}{486}$$

then one day work of (B + C) $= \frac{1}{162}$

$$B + \frac{1}{486} = \frac{1}{162}$$

$$B = \frac{1}{162} - \frac{1}{486}$$

$$B = \frac{3-1}{486} = \frac{1}{243}$$

Hence, B will complete that work alone in 243 days.

112. A can do a work in 10 day, B in 15 days and C in 20 days. A and B worked together for 4 days and then C replaced A. In how many days the entire work was finished?

(a) 16 days (b) 48/7 days
(c) 42/7 days (d) 18/7 days

RRB NTPC 16.04.2016 Shift : 2

Ans : (b) 1 day work of A and B $= \frac{1}{10} + \frac{1}{15} = \frac{1}{6}$ part

$\therefore$ 4 days work of A and B $= \frac{4}{6} = \frac{2}{3}$ part

remain work $= 1 - \frac{2}{3} = \frac{1}{3}$ part

1 day work of B and C $= \frac{1}{15} + \frac{1}{20} = \frac{7}{60}$ part

$\because$ time taken to complete work of $\frac{7}{60}$ part = 1 day

$\therefore$ time taken in complete $\frac{1}{3}$ part of work

$$= 1 \times \frac{60}{7} \times \frac{1}{3} = \frac{20}{7}$$

total time $= \frac{20}{7} + 4 = \frac{48}{7}$ days

113. P, Q and R together can do a work in 4 days. P and Q alone can do the same work in 8 and 12 days respectively. Explain how many days can R do the same work alone?

(a) 18 days (b) 20 days
(c) 22 days (d) 24 days

RRB NTPC 05.04.2016 Shift : 2

Ans : (d) work done by P, Q and R in 1 day $= \frac{1}{4}$

work done by P and Q in 1 day $= \frac{1}{8} + \frac{1}{12}$

$$= \frac{3+2}{24} = \frac{5}{24}$$

work done by R in 1 day $= \frac{1}{4} - \frac{5}{24}$

$$= \frac{6-5}{24} = \frac{1}{24}$$

Hence R will complete that work in 24 days in alone.

114. Ashok and Kiran complete a work in 10 hours Kiran and Rohan complete the same work in 15 hours, while Ashok and Rohan complete it in 12 hours. How many hours will it take for Kiran to do this alone?

(a) 26 hours (b) 15 hours
(c) 24 hours (d) 30 hours

RRB Group-D – 15/11/2018 (Shift-II)

Ans : (c) work done by Ashok and Kiran in 1 hour

$$= \frac{1}{10} \quad \text{.....(i)}$$

work done by Kiran and Rohan in 1 hour $= \frac{1}{15}$ .....(ii)

work done by Ashok and Rohan in 1 hour = $\frac{1}{12}$ .....(iii)

on adding the equation (i), (ii) and (iii)

work done by 2(Ashok, Kiran, and Rohan) in 1 hour

$$= \frac{1}{10}+\frac{1}{15}+\frac{1}{12}$$

work done by Ashok, Kiran and Rohan in 1 hour

$$= \frac{12+8+10}{120\times 2}=\frac{30}{240}=\frac{1}{8}$$

so 1 hour work of Kiran

$$= \frac{1}{8}-\frac{1}{12}=\frac{3-2}{24}=\frac{1}{24}$$

So Kiran will take 24 hours to, complete the work.

115. A, B and C can finish a work alone in 8, 9 and 12 days respectively. C starts working alone and after a day B joins him. A also joins them three days after the work begins. In how many days the entire work will be finished?

(a) $3\frac{7}{29}$ (b) $3\frac{23}{29}$

(c) $4\frac{15}{23}$ (d) $1\frac{6}{23}$

Ans. (c) : According to the question,

Suppose C started work from x day by self.

$$\frac{x-3}{8}+\frac{x-1}{9}+\frac{x}{12}=1$$

$$\frac{9x-27+8x-8+6x}{72}=1$$

$23x - 35 = 72$

$23x = 72 + 35$

$23x = 107$

$$x=\frac{107}{23}$$

$$x=4\frac{15}{23}$$

Hence all work will be finish in $4\frac{15}{23}$ days

116. A, B and C together can complete a work in 10 days. A alone can complete the work in 20 days and B alone can complete the work in 30 days. How many days will C take to complete that work alone?

(a) 30 days (b) 20 days

(c) 10 days (d) 60 days

RRB Group-D – 25/10/2018 (Shift-II)

Ans : (d) 1 day work of A, B and C = $\frac{1}{10}$ part

1 day work of A and B = $\left(\frac{1}{20}+\frac{1}{30}\right)$ part

Hence 1 day work of C = $\frac{1}{10}-\left(\frac{1}{20}+\frac{1}{30}\right)$

$$=\frac{1}{10}-\left(\frac{3+2}{60}\right)=\frac{6-5}{60}=\frac{1}{60}\text{ part}$$

Hence time taken by C to finishing work alone

$$= \frac{1}{1/60}=60\text{ days}$$

117. A and B can complete a work in 28 days, B and C can complete the same work in 35 days. While C and A can complete the same work in 42 days. How many days will C take to complete this work alone?

(a) 125 (b) 120

(c) 124 (d) 122

RRB Group-D – 12/10/2018 (Shift-I)

Ans. (b) : one day work of (A + B) = $\frac{1}{28}$ part .....(i)

one day work of (B + C) = $\frac{1}{35}$ part ..... (ii)

one day work of (C + A) = $\frac{1}{42}$ part .....(iii)

on the equation (i) + equation (ii) + equation (iii)

one day work of (A+B+B+C+C+A) = $\frac{1}{28}+\frac{1}{35}+\frac{1}{42}$

one day work of 2(A+B+C) = $\frac{15+12+10}{420}$

one day work of (A+B+C) = $\frac{37}{840}$ part ..... (iv)

on subtraction of equation (iv) to equation (i)

one day work of C = $\frac{37}{840}-\frac{1}{28}=\frac{37-30}{840}$

one day work of C = $\frac{7}{840}=\frac{1}{120}$ part

Hence C will finish all work alone in 120 days.

118. A, B and C together can complete a work in 5 days. If only A and B worked, then they use to take 54 days to complete the work and if only B and C worked, they would have taken 90 days to complete the work. If B worked alone then how many days did he take to complete the work?

(a) 145 (b) 125

(c) 135 (d) 130

RRB Group-D – 01/10/2018 (Shift-I)

Ans. (c) : one day work of (A+B+C) = $\frac{1}{45}$ part .....(i)

one day work of (A+B) = $\frac{1}{54}$ part .....(ii)

one day work of (B+C) = $\frac{1}{90}$ part .....(iii)

from the equation (ii) + equation (iii)

$$A+B+B+C=\frac{1}{54}+\frac{1}{90}$$

$$B+(A+B+C)=\frac{1}{54}+\frac{1}{90}$$

from the equation (i)

$$B=\left(\frac{1}{54}+\frac{1}{90}\right)-\frac{1}{45}$$

$$B=\frac{5+3-6}{270}=\frac{2}{270}=\frac{1}{135}\text{ part}$$

Hence B will take time of 135 days to completing work alone.

119. Rathin and Bratin together can do a work in 12 days. They start working together but Rathin has to leave work 5 days before the work ends. As a result, it took a total of 15 days to finish the work. If Bratin works alone, how many days will it take to finish this work?

(a) 30 (b) 28
(c) 20 (d) 24

RRB Group-D – 01/10/2018 (Shift-III)

Ans : (a) Suppose Rathin finish the work in R days and Batin in B days.

According to the question,

$$\frac{1}{R}+\frac{1}{B}=\frac{1}{12}\text{.....(i)}$$

and

$$\frac{10}{R}+\frac{15}{B}=1\text{ .....(ii)}$$

on doing equation (i) × 10 – equation (ii)

$$\frac{10}{B}-\frac{15}{B}=\frac{10}{12}-1$$

$$\Rightarrow\frac{-5}{B}=\frac{-2}{12}$$

B = 30 day

120. A can do a work in 10 days and B can do the same work in 20 days. With the help of C they finish the same work in 5 days. In how many days will C do the same work alone?

(a) 30 days (b) 20 days
(c) 25 days (d) 18 days

RRB NTPC 03.04.2016 Shift : 3

Ans :(b) Suppose C will do that work in x days alone

work done by A in 1 day $=\frac{1}{10}$

work done by B in 1 day $=\frac{1}{20}$

work done by C in 1 day $=\frac{1}{x}$

According to the question,

$$\frac{1}{10}+\frac{1}{20}+\frac{1}{x}=\frac{1}{5}$$

$$\frac{1}{x}=\frac{1}{5}-\frac{1}{10}-\frac{1}{20}$$

$$\frac{1}{x}=\frac{4-2-1}{20}$$

$$\frac{1}{x}=\frac{1}{20}$$

x = 20 days

121. A and B can finish a work in 10 days, B and C can finish the same work in 15 days and A and C can finish the same work in 20 days. In how many days can B finish this work alone?

(a) $\frac{20}{7}$ (b) $\frac{24}{7}$
(c) $\frac{120}{7}$ (d) $\frac{60}{7}$

RRB NTPC 12.04.2016 Shift : 1

Ans : (c) work done by (A+B+C) in one day.

$$=\frac{1}{2}\left(\frac{1}{10}+\frac{1}{15}+\frac{1}{20}\right)=\frac{1}{2}\times\left(\frac{6+4+3}{60}\right)=\frac{13}{120}$$

∴ one day work of B

= one day work of (A+B+C) – one day work of (A+C)

$$=\frac{13}{120}-\frac{1}{20}$$

$$=\frac{13-6}{120}$$

$$=\frac{7}{120}$$

Hence B will finish work in $\frac{120}{7}$ days alone.

122. A, B and C can complete a work together in 10 days. A and B would need 12 days if only the two of them worked, while B and C would take 20 days if only the two of them worked. If B worked alone, how many days would she take to complete the work?

(a) 30 (b) 22
(c) 45 (d) 20

RRB ALP & Tec. (14-08-18 Shift-II)

Ans : (a) one day work of (A+B+C) $=\frac{1}{10}$ ..... (i)

one day work of (A + B) $=\frac{1}{12}$ ..... (ii)

one day work of (B + C) $=\frac{1}{20}$ ..... (iii)

from equation (ii) + equation (iii)

$$B+(A+B+C)=\frac{1}{12}+\frac{1}{20}$$

$$\Rightarrow\quad B+\frac{1}{10}=\frac{1}{12}+\frac{1}{20}\qquad\text{\{from equation (i)\}}$$

$$\Rightarrow\quad B=\frac{1}{12}+\frac{1}{20}-\frac{1}{10}$$

$$\Rightarrow\quad B=\left(\frac{5+3-6}{60}\right)$$

$$\Rightarrow\quad B=\frac{2}{60}$$

$$\Rightarrow\quad B=\frac{1}{30}$$

Hence B will complete work in 30 days.

Type - 8

123. Charlie and Lola can do a piece of work in 20 days. They worked for 15 days, then Lola left the work. Charlie finished the remaining work in 12 days. Find out how many days Charlie can complete the work alone?

(a) 32 (b) 5
(c) 48 (d) 17

RRB Group-D – 05/11/2018 (Shift-II)

Ans : (c) Work done by Charlie and Lola in 15 days.

$= \frac{15}{20} = \frac{3}{4}$

remain work $= 1 - \frac{3}{4} = \frac{1}{4}$

Time taken by Charlie in complete $\frac{1}{4}$ of work $= 12$

Time taken by Charlie in complete work alone $= 4 \times 12$
$= 48$days

Type - 9

124. 6 boys and 10 girls can do a job in 25 days. How many days will 12 boys and 30 girls take to do the same work?

(a) 10 (b) 5
(c) 15 (d) 20

RRB RPF SI – 11/01/2019 (Shift-II)

Ans : (b) $\because$ 6B = 10 G
$\therefore$ 12B = 20G

we know that;
Formula– $M_1D_1 = M_2D_2$

$\Rightarrow 10G \times 25 = (12B + 30G) \times D_2$
$\Rightarrow 10G \times 25 = (20\,G + 30G) \times D_2$
$\Rightarrow 10 \times 25 = 50 \times D_2$
$\Rightarrow D_2 = \frac{10 \times 25}{50} = 5$ days

125. If 200 men can build a building in 1024 days then how many men will be required to construct the same building in 256 days?

(a) 650 (b) 1200
(c) 800 (d) 1400

RRB RPF Constable – 22/01/2019 (Shift-II)

Ans : (c) M = male
formula-$M_1D_1 = M_2D_2$ D = days
$200 \times 1024 = 256 \times M_2$
$M_2 = 800$

126. If 200 men, working 8 hours a day, can dig a canal in 6 days, then how long will it take for 300 men to dig the canal after working 6 hours per day?

(a) 7 days (b) $4\frac{2}{3}$ days
(c) $5\frac{1}{3}$ days (d) 6 days

RRB JE - 23/05/2019 (Shift-I)

Ans : (c) given that,
$M_1 = 200, H_1 = 8, D_1 = 6$
$M_2 = 300, H_2 = 6, D_2 = ?$
we know that;
formula
$M_1D_1H_1 = M_2D_2H_2$
$200 \times 8 \times 6 = 300 \times 6 \times D_2$
$D_2 = \frac{16}{3}$, $D_2 = 5\frac{1}{3}$ days

127. Adequate food supplies were arranged for 250 soldiers for 30 days. If 50 soldiers did not participate in the camp, how many more days would the food supply last?

(a) 10 days (b) 9 days
(c) 12 days (d) 7.5 days

RRB JE - 24/05/2019 (Shift-I)

Ans : (d) From the rule of proportion,
$M_1D_1 = M_2D_2$
$250 \times 30 = 200 \times D_2$
$\therefore D_2 = 37.5$ day
Hence food items will last more than = 37.5 – 30 = 7.5 days

128. 16 women working 7 hours per day to complete a work in 48 days. How many women will be required to complete the same work in 32 days, working 12 hours per day?

(a) 12 (b) 17
(c) 14 (d) 10

RRB JE - 25/05/2019 (Shift-II)

Ans : (c) $M_1H_1D_1 = M_2H_2D_2$
$16 \times 7 \times 48 = M_2 \times 12 \times 32$
$M_2 = 14$
Hence 14 women will be required to finish work in 32 days.

129. If 36 men can do a work in 25 days, then in how many days 15 men will do the same work?

(a) 60 days (b) 50 days
(c) 56 days (d) 54 days

RRB JE - 26/05/2019 (Shift-II)

Ans : (a) M $\rightarrow$ male
D $\rightarrow$ day
we know that
$M_1 D_1 = M_2 \times D_2$
$36 \times 25 = 15 \times x$
$x = \frac{36 \times 25}{15} = 60$ days

130. If 9 men and 12 boys can complete a work in 4 days and 4 men and 16 boys can complete the same work in 6 days, then how much time will it take for 6 men and 24 boys to complete the same work?

(a) 6 days (b) 7 days
(c) 4 days (d) 5 days

RRB JE - 27/05/2019 (Shift-II)

Ans : (c) M = male
B = boys
(9M + 12B) × 4 = (4M + 16B) × 6
(9M + 12B) × 2 = (4M + 16B) × 3
18M + 24 B = 12M + 48B
18M – 12M = 48B – 24B
6M = 24 B

$$\boxed{\frac{M}{B}=\frac{4}{1}}$$

Total work = (9M + 12B) × 4
= (36 + 12) × 4
= 48 × 4 = 192
(6M + 24B) × days = 192
(24 + 24) × days = 192

$$\text{days}=\frac{192}{24+24}=\frac{192}{48}=4$$

131. 30 people can complete a work in 12 days. How long will it take for 40 people to complete the same work?
(a) 10 (b) 9
(c) 6 (d) 8

RRB Group-D – 03/10/2018 (Shift-III)

Ans : (b) $M_1D_1 = M_2D_2$
$30 \times 12 = 40 \times D_2$

$$D_2=\frac{30\times 12}{40},\quad D_2 = 9 \text{ days}$$

9 days will be taken by 40 peoples to complete that work.

132. 45 people can make 40,000 bulbs in 12 days. How many additional people will be required to hire 100000 bulbs in 9 days?
(a) 100 (b) 105
(c) 110 (d) 120

RRB Group-D – 16/10/2018 (Shift-I)

Ans. (b) : Let x additional people will be required to making 100000 bulbs in 9 days.

formula $\frac{M_1D_1}{W_1}=\frac{M_2D_2}{W_2}$

$M_1 = 45$ $M_2 = (45+x)$
$D_1 = 12$ $D_2 = 9$
$W_1 = 40,000$ $W_2 = 100,000$

$$\frac{45\times 12}{40,000}=\frac{(45+x)\times 9}{100,000}$$

$150 = 45 + x$
$\boxed{x = 105}$
Hence 105 additional peoples will be required.

133. In a company 12 employees can make 111 candles in a given time. How many people will have to be employed to make 148 candles at the same time?
(a) 18 persons (b) 12 persons
(c) 16 persons (d) 10 persons

RRB Group-D – 31/10/2018 (Shift-II)

Ans : (c) Formula

$$\frac{M_1}{W_1}=\frac{M_2}{W_2}$$

$$\frac{12}{111}=\frac{M_2}{148}$$

$$M_2=\frac{12\times 148}{111}=16 \text{ Persons}$$

134. 34 men can do a work in 12 days. How many days will 51 men take to do this?
(a) 10 (b) 5
(c) 8 (d) 6

RRB Group-D – 04/10/2018 (Shift-II)

Ans : (c) Suppose work will be completed in x days
34 × 12 = 51 × x
x = 8 day

135. 69 people can complete a work in 6 days. How long will it take 23 people to complete the same work?
(a) 15 (b) 10
(c) 16 (d) 18

RRB Group-D – 05/10/2018 (Shift-I)

Ans. (d) : Suppose it will take x days to complete work. then,
$m_1d_1 = m_2d_2$
$69\times 6 = 23\times x$

$$\Rightarrow x=\frac{69\times 6}{23}$$

$=3\times 6 = 18$
Hence it will take 18 days to complete work of 23 people.

136. 63 people can dig a pond in 126 days. In how many days can 98 people dig it?
(a) 77 (b) 80
(c) 81 (d) 90

RRB Group-D – 11/12/2018 (Shift-II)

Ans : (c) $M_1D_1 = M_2D_2$
$63 \times 126 = 98 \times D_2$

$$\frac{63\times 126}{98}=D_2$$

$$D_2=\frac{63\times 63}{49}=81 \text{ days}$$

137. If 20 people can complete a work in 25 days then how many people will finish that work in 10 days?
(a) 45 (b) 6
(c) 100 (d) 50

RRB Group-D – 24/10/2018 (Shift-III)

Ans. (d) : Given-
$m_1 = 20$ person, $d_1 = 25$ days
$d_2 = 10$ days, $m_2 = ?$
$m_1d_1 = m_2d_2$
$20 \times 25 = 10 \times m_2$
$m_2 = 50$ persons
Hence 50 persons will complete of work in 10 days.

138. If 8 typists typing 6 hours a day, take 15 days to type a manuscript of a book, how many days will 9 typists typing 8 hours per day take to complete the same work?

(a) 10 (b) 9
(c) 11 (d) 7

RRB Group-D – 23/10/2018 (Shift-II)

Ans. (a) : Suppose number of intended days = x day
So, $M_1D_1H_1 = M_2D_2H_2$
$8 \times 15 \times 6 = 9 \times x \times 8$
$x = \frac{15 \times 6}{9} = \frac{90}{9}$
x = 10 days

139. A hostel has a stock of 6,190.80 Kg of wheat for 105 students for 22 days. After 5 days, 14 more students come to the hostel. It all the students eat the same food, how much remaining wheat will be enough for the students?

(a) 15 days (b) 11 days
(c) 1 days (d) 17 days

RRB Group-D – 04/10/2018 (Shift-I)

Ans. (a) Suppose remain wheat will be enough for x days.

$$\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$
$$\frac{105 \times 22}{6190.80} = \frac{105 \times 5}{6190.80} + \frac{(105+14)x}{6190.80}$$
$105 \times 22 = 105 \times 5 + (105 + 14)\,x$
$105 \times 22 = 105 \times 5 + 119x$
$2310 = 525 + 119x$
$2310 - 525 = 119x$
$\frac{1785}{119} = x$
$\boxed{x = 15}$ days

140. If 15 boys earn Rs. 750 in 5 days, then how much money 25 boys will earn in 6 days?

(a) Rs. 1,500 (b) Rs. 960
(c) Rs. 1,200 (d) Rs. 900

RRB Group-D – 18/09/2018 (Shift-I)

Ans. (a) M = Male, D = day, W = work

$$\frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$
$$\Rightarrow \frac{15 \times 5}{750} = \frac{25 \times 6}{W_2}$$
$$W_2 = \frac{25 \times 6 \times 750}{75} = \text{Rs. } 1500$$

141. If 2 men or 3 women can complete a work in 30 days, then in how many days will 6 men and 1 woman be able to complete the same work?

(a) 4 days (b) 6 days
(c) 5 days (d) 9 days

RRB Group-D – 18/09/2018 (Shift-III)

Ans. (d) : According to the question,
(M = Male W = Woman)
$2M \times 30 = 3W \times 30$
$2M = 3W$
$1M = \frac{3}{2}W$
Total women $6M + 1W = 6 \times \frac{3}{2}W + 1W = 10W$
$M_1D_1 = M_2D_2$
$3 \times 30 = 10 \times D_2$
$\therefore D_2 = \frac{3 \times 30}{10} = 9$ days

142. 16 men make a model in 35 days. How long will it take 25 men to make this model?

(a) 22.2 (b) 22.1
(c) 22 (d) 22.4

RRB NTPC 17.01.2017 Shift-3

Ans : (d) $M_1 \times D_1 = M_2 \times D_2$
$16 \times 35 = 25 \times D_2$
$D_2 = \frac{16 \times 35}{25} \Rightarrow \boxed{D_2 = 22.4}$
So, It will take 22.4 days for 25 mens to model.

143. If 5 men can do a work in 9 days, then in how many days will 3 men finish the same work?

(a) 12 (b) 15
(c) 13 (d) 18

RRB NTPC 05.04.2016 Shift-1

Ans : (b) $\because M_1 D_1 = M_2 D_2$
$\therefore 5 \times 9 = 3 \times D_2$
$\Rightarrow D_2 = \frac{5 \times 9}{3}$
= 15 days

144. 30 men can do a work in 16 days by working 5 hours per day. In how many days will 40 men complete the same work by working 6 hours per day?

(a) in 12 days (b) in 10 days
(c) in 15 days (d) in 18 days

RRB NTPC 02.04.2016 Shift : 2

Ans : (b) M_1 = 30 male M_2 = 40 male
H_1 = 5 hours, H_2 = 6 hours
D_1 = 16 days, D_2 = ?
$D_2 = \frac{M_1D_1H_1}{M_2H_2}$
$D_2 = \frac{30 \times 16 \times 5}{40 \times 6}$
D_2 = 10 days

145. 10 men and 5 women complete a work in 60 days. If a man can do the work of two women, then how much time will 5 men and 20 women take to complete half of that work?

(a) 25 (b) 36
(c) 27 (d) 50

RRB NTPC 18.01.2017 Shift : 1

Ans : (a) $\because \frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$

According to the question,

$$\frac{(10M+5W)60}{1} = \frac{(5M+20W)\times D_2}{\frac{1}{2}} \text{.....(i)}$$

$\because$ 1M = 2W (given)

on putting 1M = 2W in equation (1)

$$\frac{(10\times 2W+5W)\times 60}{1} = \frac{(5\times 2W+20W)2\times D_2}{1}$$

$25W\times 60 = 30W \times 2 \times D_2$

$D_2 = 25$ days

146. Girish can paint a picture in 4 days if he works for 8 hours per day. So due to illness he can work only for 6 hours a day and he has to complete 7 other such pictures. How many days will Girish take to complete the work?

(a) 28 (b) $37\frac{1}{3}$
(c) $39\frac{2}{5}$ (d) 32

RRB NTPC 18.01.2017 Shift : 2

Ans :(b) given-

$M_1 = 1, M_2 = 1$
$D_1 = 4, D_2 = ?$
$H_1 = 8, H_2 = 6$
$W_1 = 1\ W_2 = 7$

According to the question,

$$\frac{M_1D_1H_1}{W_1} = \frac{M_2D_2H_2}{W_2}$$

$$\frac{1\times 4\times 8}{1} = \frac{1\times D_2\times 6}{7}$$

$$D_2 = \frac{32\times 7}{6}$$

$$\boxed{D_2 = \frac{112}{3} = 37\frac{1}{3}\text{days}}$$

147. In a factory, 60 workers can sew 120 meters of clothes in 7 days, how many long clothes will be stitched in 5 days by 70 workers?

(a) 100 m (b) 90 m
(c) 85 m (d) 110 m

RRB NTPC 19.04.2016 Shift : 2

Ans : (a) Given that,

$M_1 = 60$ manual worker, $D_1 = 7$ days, $W_1 = 120$ meter
$M_2 = 70$ manual worker, $D_2 = 5$ days, $W_2 = ?$

$$\because \frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}$$

$$\therefore \frac{60\times 7}{120} = \frac{70\times 5}{W_2} \Rightarrow \frac{7}{2} = \frac{350}{W_2}$$

$W_2 = 100$ meter

148. 12 people can complete a work in X days, now 8 extra people are employed. The entire work was completed in 60 days. Find the value of X.

(a) 80 (b) 100
(c) 55 (d) 45

RRB NTPC 18.04.2016 Shift : 3

Ans : (b) Given-

$M_1 = 12$ males, $D_1 = x$ day
$M_2 = 12 + 8 = 20$ males, $D_2 = 60$ day
$M_1D_1 = M_2D_2$
$12 \times x = 20 \times 60$

$$x = \frac{20\times 60}{12}$$

$x = 100$ days

149. In a project a team of 54 members can do a work in 35 hours. In how many hours can 18 members do the same work?

(a) 90 (b) 120
(c) 105 (d) 110

RRB NTPC 12.04.2016 Shift : 2

Ans : (c) $M_1 = 54$ members
$H_1 = 35$ hours
$M_2 = 18$ members
$H_2 = ?$
we know that;

$$H_2 = \frac{M_1\times H_1}{M_2}$$

$$= \frac{54\times 35}{18}$$

$= 105$ hours

150. If 18 men construct a model of a ship in 7 days, then how much time will it take to prepare this model by 15 men?

(a) 8.7 (b) 8.5
(c) 8.4 (d) 43/5

RRB NTPC 19.01.2017 Shift : 3

Ans : (c) $M_1D_1 = M_2D_2$

$\Rightarrow 18 \times 7 = 15 \times D_2$

$$D_2 = \frac{126}{15} = 8.4$$

151. A piece of work can be done by 16 men in 8 days working 12 hours a day. How many men are needed to complete another work, which is three times the first one, in 24 days working 8 hours a day ?

(a) 22 male (b) 23 male
(c) 25 male (d) 24 male

RRB ALP & Tec. (17-08-18 Shift-I)

Ans : (d) Suppose to finish the work x men are required

$$\frac{m_1h_1d_1}{w_1} = \frac{m_2h_2d_2}{w_2}$$

According to the question,

$$\frac{16\times 8\times 12}{1} = \frac{x\times 8\times 24}{3}$$

$x = 24$

So number of intended persons = 24 males

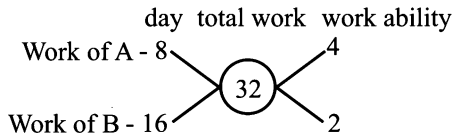
Type - 10

152. P and Q can do any work separately in 8 and 16 days respectively. They worked together and got the job done when the work is done, what is the ratio of the work done by P and Q

(a) 2 : 5 (b) 3 : 2
(c) 2 : 1 (d) 2 : 3

RRB JE - 23/05/2019 (Shift-II)

Ans : (c)



∴ A : B = 4 : 2 = 2 : 1

153. 20 women can do in 16 days same work completed by 16 men in 15 days. Find the ratio of capacity of a man to a woman.

(a) 5:6 (b) 3:4
(c) 6:7 (d) 4:3

RRB JE - 26/05/2019 (Shift-III)

Ans : (d) ∵ $M_1 D_1 = M_2 D_2$

$20W \times 16 = 16M \times 15$

$\frac{1M}{1W} = \frac{20\times16}{16\times15} = \frac{4}{3}$

M : W = 4 : 3

154. 6 men and 2 boys can do five times more work than a man. Find the ratio of work capacity of a man to that of a boy.

(a) 3 : 1 (b) 4 : 1
(c) 1 : 3 (d) 3 : 2

RRB JE - 27/05/2019 (Shift-I)

Ans : (a) According to the question,

(6M + 2B) = (1M + 1B) 5 {where M→ male B→ boy}

6M + 2B = 5M + 5B

1M = 3B

M : B = 3 : 1

155. If 4 boys and 6 girls can do a work in 8 days and 3 boys and 7 girls can do it in 10 days. Find the ratio of work capacity of a boy to a girl.

(a) 5:12 (b) 11:1
(c) 12:5 (d) 1:11

RRB JE - 27/05/2019 (Shift-III)

Ans : (b) According to the question,

(4 boys + 6 girls) 8 = (3 boys + 7 girls) 10

32 boys + 48 girls = 30 boys + 70 girls

2 boys = 22 girls

$\frac{1 \text{ boys}}{1 \text{ girls}} = \frac{22}{2} = \frac{11}{1} = 11:1$

156. If 6 men and 8 boys can complete a work in 10 days and 26 men and 48 boys can do it in 2 days, then find the ratio of the capacity of a man to the capacity of a boy.

(a) 4 : 3 (b) 2 : 1
(c) 6 : 5 (d) 3 ; 2

RRB JE - 28/05/2019 (Shift-II)

Ans : (b) Formula- $M_1D_1 = M_2D_2$

(6 males + 8 boys)10 = (26 males + 48 boys)2

60 males + 80 boys = 52 males + 96 boys

60 males − 52 males = 96 boys − 80 boys

8 males = 16 boys

$\frac{\text{males}}{\text{boys}} = \frac{16}{8} = \frac{2}{1}$

male : boys = 2 : 1

157. If 12 men and 6 boys can do a work in 4 days and 4 men and 14 boys can do the same work in 8 days, then find the ratio of the capacity of a man and a boy.

(a) 2:11 (b) 11:2
(c) 3:7 (d) 2:5

RRB RPF Constable – 24/01/2019 (Shift-III)

Ans. (b) According to question, (M = male B = boys)

(12 M + 6 B) × 4 = (4 M + 14 B) × 8

12 M + 6 B = 8 M + 28 B

4 M = 22 B

2 M = 11 B

M : B = 11 : 2

158. P can do 2/5 part of a work in 10 days. If P and Q can do 1/3 part of the same work in 5 days, find the ratio of their work capacity.

(a) 3 : 4 (b) 4 : 3
(c) 3 : 2 (d) 2 : 3

RRB JE - 27/06/2019 (Shift-III)

Ans : (c) P, 2/5 work = 10 day

then P will complete work = $10\times\frac{5}{2}$ = 25 day

from the question-

$\frac{(P+Q)5}{\frac{1}{3}} = \frac{P\times25}{1}$ $\left[\because \frac{M_1D_1}{W_1} = \frac{M_2D_2}{W_2}\right]$

15P + 15Q = 25 P

15 Q = 10P

$\frac{P}{Q} = \frac{15}{10}$

$\frac{P}{Q} = \frac{3}{2}$

ratio of work ability = 3 : 2

159. In a village Under Mission Kakatiya, 38 men working 6 hours a day can do a piece of work in 12 days. Find the number of days in which 57 men working in 8 hours a day can do twice the work. Assume that 2 men of the first group do as much work in 1 hours as 3 men of the second group do in $11\frac{1}{2}$.

(a) 27 days (b) 28 days
(c) 24 days (d) 32 days

RRB Group-D – 12/10/2018 (Shift-III)

Ans : (a) Work ability ratio of both group of males

$$= 3\times1\frac{1}{2} : 2\times1 \quad = \frac{9}{2} : 2 \quad = 9:4$$

formula

$$\frac{M_1D_1H_1P_1}{W_1} = \frac{M_2D_2H_2P_2}{W_2}$$

$$\frac{38\times12\times6\times9}{1} = \frac{57\times D_2\times8\times4}{2}$$

$D_2 = 27$ days

Therefore, In 27 days 57 males can do work doubled to 8 hours work

160. The working efficiency ratio of P and Q is 5:7 find the ratio of days taken by them to finish a work.

(a) 7:5 (b) 3:4
(c) 4:3 (d) 5:7

RRB NTPC 06.04.2016 Shift : 2

Ans : (a) Working ratio of P and Q = 5:7

ratio of days $= \frac{1}{5} : \frac{1}{7}$ $\left(\because \text{time} \propto \frac{1}{\text{efficiency}}\right)$

$= 7 : 5$

161. Adil and Viren together can complete a task in 20 days. Biren and Chirag together can complete the same task in 50 days. Adil and Chirag together can complete the same task in 40 days. If the same task was to be done by them alone, what would be the ratio of time taken by Adil alone to taken by Viren?

(a) 11 : 9 (b) 11 : 3
(c) 7 : 9 (d) 9 : 11

RRB NTPC 28.04.2016 Shift : 2

Ans : (d)

one day work of (Adil + Viren) $= \frac{1}{20}$ part .....(i)

one day work of (Viren + Chirag) $= \frac{1}{50}$ part .....(ii)

one day work of (Adil + Chirag) $= \frac{1}{40}$ part .....(iii)

on adding of equation (i), (ii) and (iii)
one day work of 2(Adil + Viren + Chirag)

$$= \frac{1}{20} + \frac{1}{50} + \frac{1}{40}$$

$$= \frac{10+4+5}{200} = \frac{19}{200} \text{ part}$$

$\therefore$ one day work of Adil $= \frac{19}{400} - \frac{1}{50}$

$$= \frac{19-8}{400} = \frac{11}{400}$$

$\therefore$ Adil time taken alone to complete work $= \frac{400}{11}$ days

now 1 day work of Viren $= \frac{19}{400} - \frac{1}{40}$

$$= \frac{19-10}{400} = \frac{9}{400}$$

$\therefore$ Time taken by Viren to complete work

$= \frac{400}{9}$ days

$\therefore$ initended ratio $= \frac{400}{11} : \frac{400}{9} = 9 : 11$

162. A and B together can do a work in 40 days. The ratio of their working rate is 8:5 In how many days will A alone complete the same work.

(a) 65 days (b) 40 days
(c) 72 days (d) 104 days

RRB NTPC 02.04.2016 Shift : 1
RRB NTPC 29.04.2016 Shift : 3

Ans : (a) Let A and B complete work separately in 5x and 8x days

$\therefore$ work done by both in 1 day $= \frac{1}{40}$ part

According to the question,

$$\frac{1}{5x} + \frac{1}{8x} = \frac{1}{40}$$

$$\frac{13}{40x} = \frac{1}{40}$$

$$x = 13$$

$\therefore$ A will complete that work alone in 65 days.

163. P can finish a work in 60 days and Q can finish the same work in 50 days. Find the ratio of the working capacities of P and Q.

(a) 1/3 (b) 6/5
(c) 5/6 (d) 4/5

RRB NTPC 19.04.2016 Shift : 2

Ans : (c) work ability $\propto \frac{1}{\text{time}}$

work ability of P $= \frac{1}{60}$

work ability of Q $= \frac{1}{50}$

So work's ability ratio of P and Q $= \frac{\frac{1}{60}}{\frac{1}{50}}$

$$= \frac{50}{60}, \quad = \frac{5}{6}$$

164. Five women can do a work in thirty six days. If the ratio between the capacity of a man and a woman is 3:1, then find how many days it will take 5 men to complete the same work?

(a) 12 days (b) 15 days
(c) 18 days (d) 108 days

RRB NTPC 18.04.2016 Shift : 1

Ans : (a) Given-

	man		women
working ability	3	:	1

$\left\{\because \text{work ability} \propto \frac{1}{\text{time}}\right\}$

$\therefore$ time = 1 : 3

$\because$ 5 women does works in 36 days.

$\therefore$ time taken by 5 man in doing work $= 36 \times \frac{1}{3}$

$= 12$ days

165. Carpenters A and B work together then they can complete a work in 10 days. Carpenter A is twice faster than carpenter B. If carpenter B works alone, how long will he take to complete this work?

(a) 30 (b) 15
(c) 20 (d) 10

RRB NTPC 29.04.2016 Shift : 1

Ans : (a) Ability ratio of carpenter A and carpenter B= 2:1

$\therefore$ time ratio = 1: 2

According to the question,

$$\frac{1}{x}+\frac{1}{2x}=\frac{1}{10} \qquad \left[\text{time}=\frac{1}{\text{ability}}\right]$$

$$\Rightarrow \frac{2+1}{2x}=\frac{1}{10}$$

$$\frac{3}{2x}=\frac{1}{10}$$

$2x = 30$

$x = 15$

taken day by A = x = 15 day

taken day by B = 2x = 15 × 2 = 30 day

166. Together Rahul and Raghav can pluck 260 flowers in 1 hour. Their flower breaking abilities are in the ration of 8:5 Find the number of flowers to be plucked by Raghav.

(a) 100 (b) 130
(c) 78 (d) 80

RRB NTPC 30.04.2016 Shift : 1

Ans : (a) Ability ratio of Rahul and Raghav = 8:5

Number of plucking flowers by Raghav

$$= 260 \times \frac{5}{13} = 100$$

Type - 8

167. 2/5 of a set of notebooks are sold on the first day. 3/4 of the remaining were sold on the second day. If there are still 75 notebooks left, how many notebook were kept for sale.

(a) 1000 (b) 500
(c) 250 (d) 750

RRB RPF SI – 16/01/2019 (Shift-II)

Ans. (b) Let total notebook were kept 100 for sell

According to the question,

sold notebook in first day $= 100 \times \frac{2}{5} = 40$

left notebook = 100 – 40 = 60

sold notebook in second day $= 60 \times \frac{3}{4} = 45$

left notebook = 100 – (40 + 45)

75 = 100 – 85 = 15

15 = 75

$$100 = \frac{75}{15} \times 100 = 500$$

Hence number of total notebooks = 500

168. Amit packs 20 chocolates in 40 minutes. Guneet packs 30 chocolates in 45 minutes. How much chocolate can both pack together in 1 hours?

(a) 60 (b) 80
(c) 90 (d) 70

RRB Group-D – 11/10/2018 (Shift-III)

Ans : (d) Packed chocolates number by Amit in 1 minute $= \frac{20}{40} = \frac{1}{2}$

packed chocolates number by guneet in 1 minute

$$= \frac{30}{45} = \frac{2}{3}$$

packed chocolate by Amit and Guneet in 1 minute

$$= \frac{1}{2} + \frac{2}{3} = \frac{7}{6}$$

packed chocolate's number in 1 hours or 60 minutes

$$= \frac{7}{6} \times 60 = 70$$

169. If 2 minutes 30 second of internet download bill is 18 rupees, then what will be the bill of 3 minutes 20 seconds? (up to one decimal place)

(a) 24 (b) 24.1
(c) 24.2 (d) 23.9

RRB NTPC 11.04.2016 Shift : 3

Ans : (a)

2 minute 30 sec = 150 seconds bill = 18

$\therefore$ 3 minute 20 sec = 200 seconds bill $= \frac{18 \times 200}{150}$

= Rs. 24

170. The bill for a satellite airtime for 2 minute 30 seconds is Rs. 25, then what will be the price of 3 minutes 20 seconds in rupess? (up to one decimal place)

(a) 33.3 (b) 33.2
(c) 33.4 (d) 33.1

RRB NTPC 07.04.2016 Shift : 1

Ans : (a) 2 minute 30 second or (120 + 30) second's bill = Rs. 25

1 second's bill $= \frac{25}{150} = \frac{1}{6}$

then

3 minute 20 second = 180 + 20 = 200 second

200 second's bill $= \frac{1}{6} \times 200 = \frac{100}{3} = 33.33$

171. The fax bill for 3 minutes 40 seconds is Rs. 15 what will be the bill for 2 minutes 10 seconds? (up to one decimal place)

(a) 8.9 (b) 8.7
(c) 8.6 (d) 8.8

RRB NTPC 07.04.2016 Shift : 2

Ans : (d) 3 minutes 40 seconds = 3 × 60 + 40
= 220 seconds

2 minutes 10 seconds = 2 × 60 + 10
= 130 seconds

∵ 220 second's fax bill = Rs. 15

∴ 130 second's fax bill $= \frac{15}{220} \times 130$

$= \frac{195}{22}$ = Rs. 8.8

172. The bill for 3 minutes 20 seconds telephonic conversation is Rs. 35.50. What is the bill for a call of 5 minutes and 30 seconds? (rounded to one decimal place)

(a) 58.6 (b) 58.7
(c) 58.5 (d) 58.8

RRB NTPC 19.01.2017 Shift : 3

Ans : (a) Telephone converstation's bill in 3 minute 20 second

$= \frac{10}{3}$ minute = Rs. 35.50

telephone conversation's bill in 1 minute $= \frac{35.50 \times 3}{10}$

telephone conversation's bill in 5 minute 30 second

$= \frac{35.50 \times 3}{10} \times \frac{11}{2}$

= 58.575 = Rs. 58.6

173. The bill is 13 rupees for 5 minutes 40 seconds of satellite airtime. How much is the bill for 2 minutes 40 seconds? (to one decimal place)

(a) 6.1 (b) 6
(c) 6.2 (d) 6.3

RRB NTPC 26.04.2016 Shift : 2

Ans : (a) 5 minutes 40 seconds

$= 5 \times 60 + 40 = 340$ seconds

340 second's bill = 13 Rs.

1 second's bill $= \frac{13}{340}$

2 minutes 40 seconds (120 + 40 = 160 second's) bill

$= \frac{13 \times 160}{340}$ = Rs. 6.1

174. A school canteen buys 3200 kg of pulses which will be enough to provide 20 days nutrition to 1200 children. 6 days later 180 children went on a month long journey. How many more days of pulses will be enough for the remaining children?

(a) 14 (b) 16.5
(c) 12.6 (d) 17.2

RRB NTPC 28.04.2016 Shift : 1

Ans :(b) Let pulses for the nutrition of the remaining children is sufficient for x days.

According to the question,

$1200 \times 14 = (1200 - 180) \times x$

$\Rightarrow 1200 \times 14 = 1020 \times x$

$\Rightarrow x = \frac{16800}{1020}$ or

$x = 16.47 \approx 16.5$ day

175. Rekha reads 15 pages in 25 minutes. How many pages will she read in 45 minutes?

(a) 30 (b) 27
(c) 25 (d) 35

RRB NTPC 19.01.2017 Shift : 2

Ans : (b)

number of pages read by Rekha in 25 minute =15

∴ number of pages read by Rekha in 45 minute

$= \frac{15 \times 45}{25} = 27$

176. Type 1 workers are 2.5 times as efficient as Type 2 workers. Twelve Type 1 workers can complete a task in 10 days. How many days would it have taken 4 Type 1 and 15 Type 2 workers to do the same work?

(a) 13 (b) 10
(c) 12 (d) 11

RRB ALP & Tec. (21-08-18 Shift-III)

Ans : (c) Let it will take x days to complete work

According to the question,

$10 \times 12 = \left(4 + \frac{15}{2.5}\right) \times x$

$10 \times 12 = \left(\frac{10+15}{2.5}\right) \times x$

$2.5 \times 10 \times 12 = 25 \times x$

$25 \times 12 = 25 \times x$

$x = 12$

Hence it will take 12 days to complete work.